**Wireless Weather Station Technical Manual**

**Release 1.0**

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**Document Control**

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**Revision History**

| Revision | Date | Description |
|---|---|---|
| 0.1 | Initial Development | Original weather station concept |
| 0.5 | Development Phase | Multi-controller architecture established |
| 0.6 | Development Phase | LoRa communications implemented |
| 0.7 | Development Phase | Solar-powered mast-mounted node added |
| 0.8 | Development Phase | Remote reboot and uptime monitoring added |
| 0.9 | Development Phase | Dashboard and database architecture expanded |
| 1.0 | October 2026 | Initial technical manual release |

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**Acronyms and Definitions**

| Acronym | Definition |
|---|---|
| ADC | Analog-to-Digital Converter |
| BMP280 | Bosch Barometric Pressure Sensor |
| DS3231 | Real-Time Clock Module |
| EEPROM | Electrically Erasable Programmable Read-Only Memory |
| ESP32 | Espressif 32-bit Microcontroller |
| GPIO | General Purpose Input / Output |
| HTTP | Hypertext Transfer Protocol |
| $I^2C$ | Inter-Integrated Circuit |
| LCD | Liquid Crystal Display |
| LoRa | Long Range Radio |
| RTC | Real-Time Clock |
| SHT31-D | Temperature/Humidity Sensor |
| UART | Universal Asynchronous Receiver / Transmitter |
| WU | Weather Underground |

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## Section 1 – Introduction

### 1.1 Purpose

This manual documents the design, architecture, operation, maintenance, and future development of the Wireless Weather Station system.

The system combines environmental sensing, local monitoring, wireless telemetry, Internet connectivity, database storage, and dashboard presentation into a fully integrated weather-monitoring platform.

The document is intended to serve as the primary engineering reference for:

- Construction
- Maintenance
- Troubleshooting
- Firmware modification

- Future expansion

---

### 1.2 System Description

The weather station is built around a distributed architecture based on three cooperating controllers:

**Arduino Mega**

The Arduino Mega functions as the primary supervisory controller.

Responsibilities include:

- User interface management
- Sensor integration
- EEPROM management
- RTC management
- Min/max tracking
- Mast-mounted node supervision
- Data aggregation

**ESP32-S3 Gateway**

The ESP32-S3 serves as the Internet gateway and presentation platform.

Responsibilities include:

- Dashboard hosting
- Wi-Fi communications
- Database uploads
- Weather Underground uploads
- Status monitoring

**ESP32-C3 Mast-mounted Node**

The ESP32-C3 serves as the mast-mounted telemetry controller.

Responsibilities include:

- Wind-speed measurement
- Wind-direction processing
- Rainfall measurement
- LoRa communications
- Mast-mounted system uptime tracking

---

### 1.3 Design Goals

The weather station was designed around several key objectives:

**Reliability**

The station must continue operating through temporary communications failures and Internet outages.

**Serviceability**

The system should be diagnosable without specialized equipment.

**Modularity**

Major subsystems must be replaceable independently.

**Expandability**

Future sensors and capabilities should be incorporated without redesigning the architecture.

**Data Integrity**

Weather observations and system status information must be preserved for analysis and long-term review.

---

### 1.4 Major Features

The current system provides:

- Temperature monitoring
- Relative humidity monitoring
- Dew-point calculation
- Barometric pressure measurement

- Wind-speed measurement
- Wind-gust measurement
- Wind-direction measurement
- Rainfall measurement
- Historical data storage
- Browser-based dashboard
- Weather Underground integration
- Automatic fault recovery
- Solar-powered mast-mounted sensor operation

---

**Section 2 – System Overview**

**2.1 System Functions**

The station performs four major functions:

Measure

Transmit

Display

Archive

Environmental data is acquired by the Radiation Shield Sensor Package and the Mast-mounted Sensor System, transmitted through communications links to the Main Controller, displayed locally and remotely, and archived within a database.

---

**2.2 Environmental Measurements**

The station currently measures:

| **Parameter** | **Source** |
|---|---|
| Temperature | SHT31-D / BMP280 |
| Humidity | SHT31-D |

| | |
|---|---|
| Dew Point | Calculated |
| Pressure | BMP280 |
| Wind Speed | Anemometer |
| Wind Gust | Anemometer |
| Wind Direction | Wind Vane |
| Rainfall Rate | Rain Gauge |
| Daily Rainfall | Rain Gauge |
| Mast-mounted Sensor Uptime | ESP32-C3 |

---

**2.3 Physical Subsystems**

The weather station is organized into three major physical assemblies:

**Main Station**

Contains the Arduino Mega, LCD interface, ESP32-S3 gateway, and indoor LoRa transceiver.

**Radiation Shield Sensor Package**

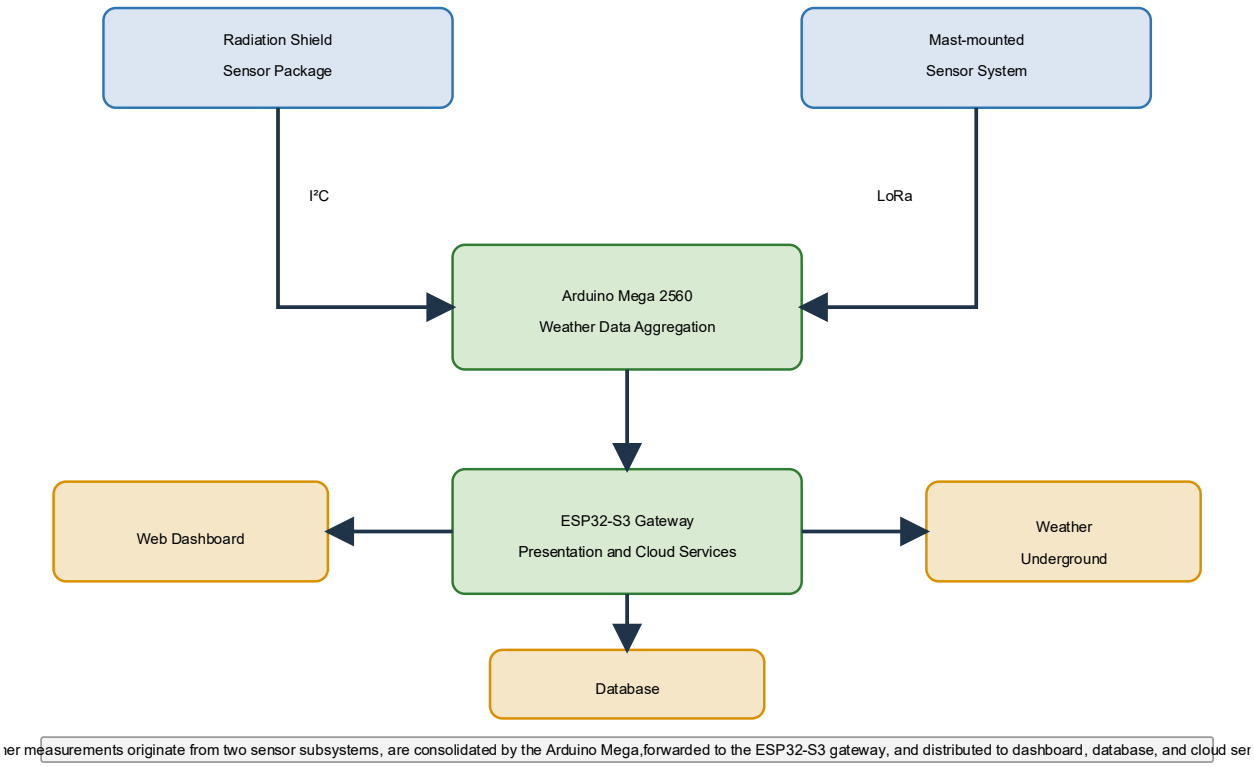
Contains the SHT31-D, BMP280, and DS3231 RTC mounted outside the office window.

**Mast-mounted Sensor System**

Contains the ESP32-C3 controller, wind vane, anemometer, rain gauge, LoRa transceiver, and solar power subsystem.

### 2.4 Data Path

Environmental measurements traverse the following path:




ıer measurements originate from two sensor subsystems, are consolidated by the Arduino Mega,forwarded to the ESP32-S3 gateway, and distributed to dashboard, database, and cloud ser

This architecture separates measurement from presentation and storage.

---

### 2.5 Operational Cycle

The normal operating cycle is:

1. Mast-mounted Sensor System acquires wind and rainfall measurements.
2. ESP32-C3 transmits telemetry to the Main Station via LoRa.
3. Radiation Shield Sensor Package acquires temperature, humidity, pressure, and time data.
4. Arduino Mega aggregates measurements from both sensor subsystems.
5. Arduino Mega generates a consolidated weather packet.
6. ESP32-S3 receives the packet.
7. Dashboard updated.
8. Database updated.

9. Weather Underground updated.

The cycle typically repeats every five seconds.

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**Section 3 – System Architecture**

**3.1 Architectural Overview**

The weather station utilizes a three-controller distributed architecture.

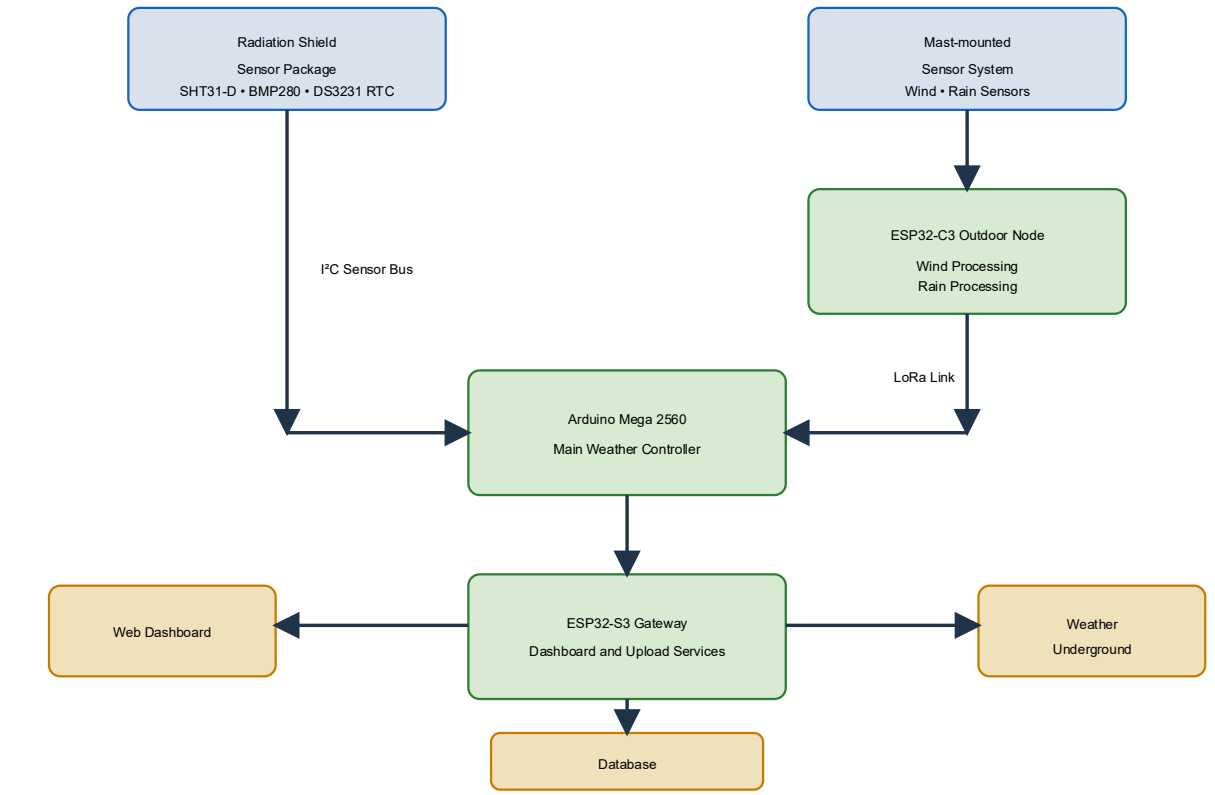
This design distributes responsibilities across specialized controllers rather than placing all functions on a single device.

Advantages include:

- Improved reliability
- Lower software complexity
- Easier debugging
- Improved scalability

---

**Figure 3-1. System Architecture Diagram**




lata originates from the Radiation Shield Sensor Package and the Mast-mounted Sensor System, is consolidated by the Arduino Mega, processed by the ESP32-S3 gateway, and distributed to local and cloud

---

## 3.2 Controller Responsibility Matrix

| Controller | Primary Responsibilities |
|---|---|
| Arduino Mega | User interface, sensing, recovery management |
| ESP32-S3 | Dashboard, Wi-Fi, uploads |
| ESP32-C3 | Mast-mounted measurements and telemetry |

---

## 3.3 Main Station Architecture

The main station controller subsystem consists of:

- Arduino Mega

- LCD Keypad Shield
- LoRa Module
- ESP32-S3

The Mega acts as the central authority for environmental processing and operational management.

---

### 3.4 Mast Controller Architecture

The mast-mounted subsystem contains:

- ESP32-C3
- LoRa module
- Wind vane
- Anemometer
- Rain gauge
- Solar power subsystem

This subsystem operates autonomously and transmits weather observations to the main station.

---

### 3.5 Fault Tolerance

The architecture contains multiple layers of fault tolerance:

**Sensor Redundancy**

Temperature can be obtained from multiple sources.

**Communications Monitoring**

LoRa communication health is continuously monitored.

**Automatic Recovery**

The Mega can command remote reboots of the mast-mounted node.

**Dashboard Visibility**

System health information is continuously available for diagnostics.

## Section 4 – Hardware Overview

### 4.1 Hardware Architecture

The weather station hardware is divided into:

Main Station

Radiation Shield Sensor Package

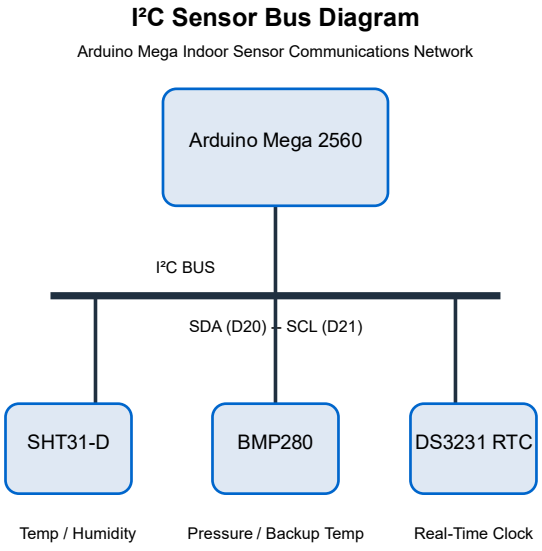
Mast-mounted Sensor System

Power System

Communications System

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**Figure 4-1. I²C Sensor Bus Diagram**

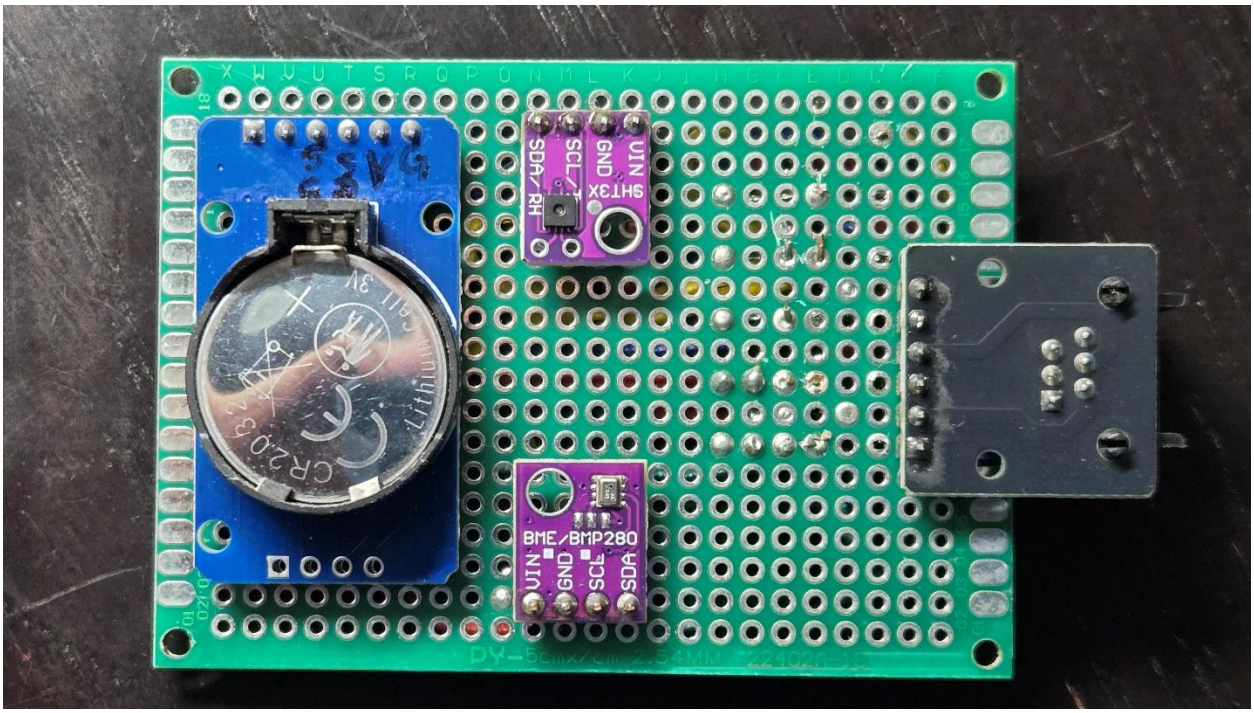



Indoor sensor communications bus showing the Arduino Mega, SHT31-D temperature/humidity sensor, BMP280 pressure sensor, and DS3231 real-time clock connected through the shared I²C interface using SDA (D20) and SCL (D21).

---

The SHT31-D, BMP280, and DS3231 RTC sensors are located on a small perfboard that is shielded from direct sun inside a radiation shield.  Here is a picture of the perfboard:

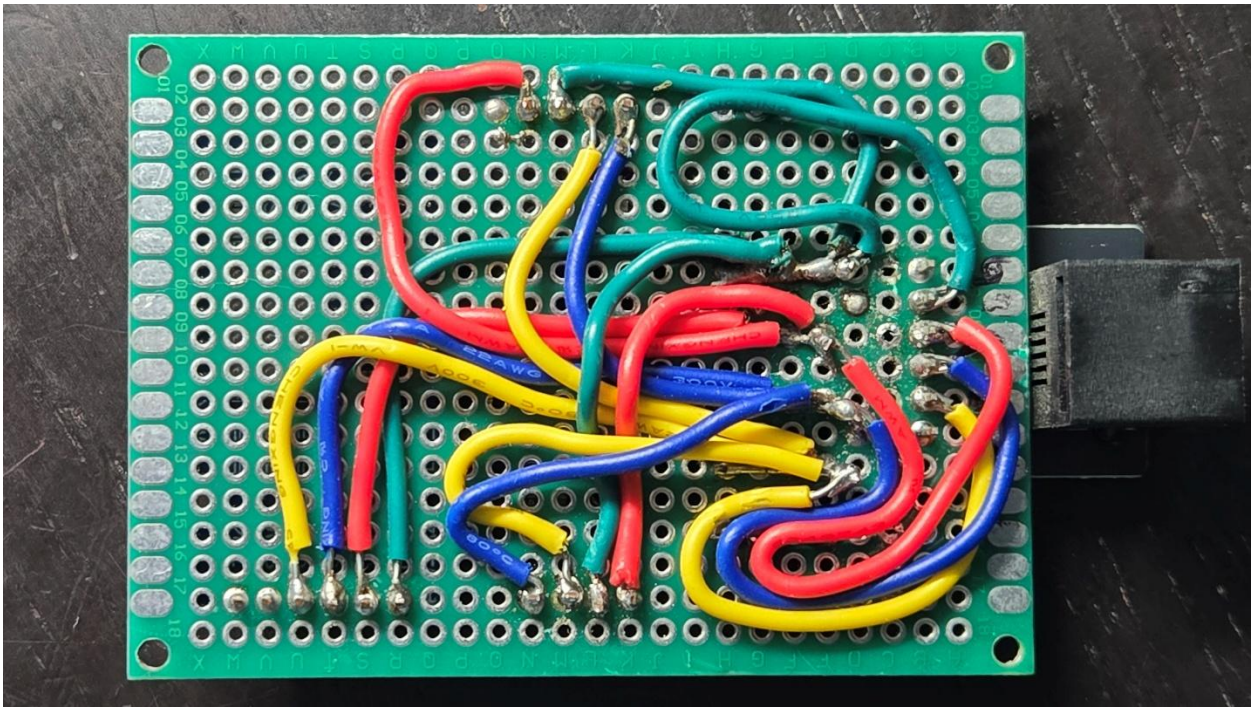
**Figure 4-2. Sensors inside Radiation Shield**




The DS3231 RTC is on the left, showing its CR2032 cell. The SHT31-D sensor, which serves as the current primary temperature and humidity sensor, is the small purple module at top middle. The BMP280 pressure sensor is the small purple module at bottom middle of the board. The RJ12 socket for the cable connector is at the right.

Here is a picture of the underside of the perfboard:

**Figure 4-3. Underside of Radiation Shield Perfboard**



The green wires are GND.

The red wires are 3.3v power.

The yellow wires are SCL ($I^2C$ clock).

The blue wires are SDA ($I^2C$ data).

**Figure 4-4. Radiation Shield Installation**



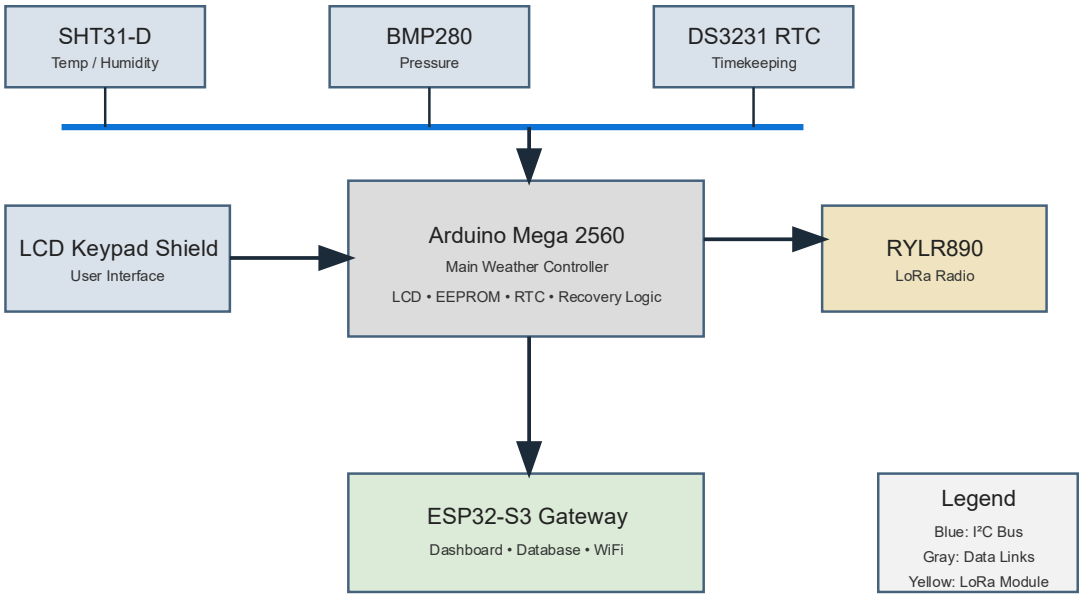
*Radiation shield outside office window, used for temperature and humidity measurements.*

### 4.2 Main Station Hardware Inventory

| Component | Function |
|---|---|
| Arduino Mega 2560 | Main controller |
| LCD Keypad Shield | Local interface |
| RYLR890 | LoRa communications |
| ESP32-S3 | Gateway controller |

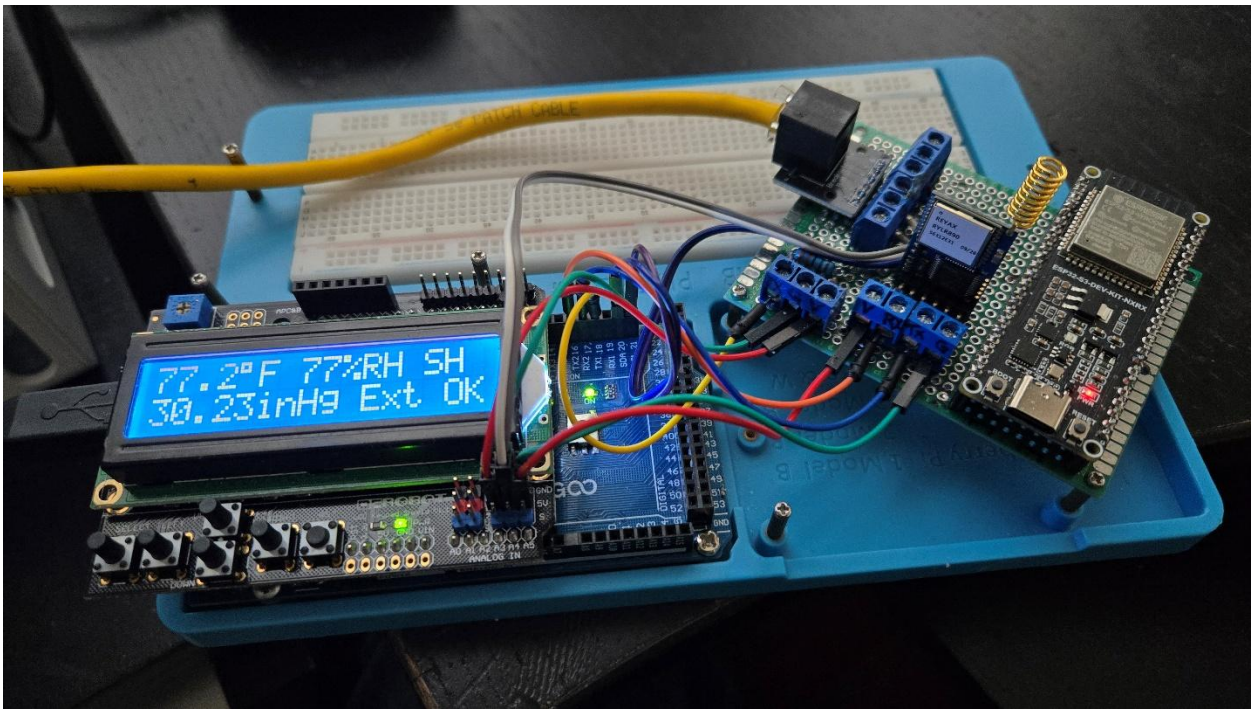
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**Figure 4-5. Main Station Electronics Diagram**



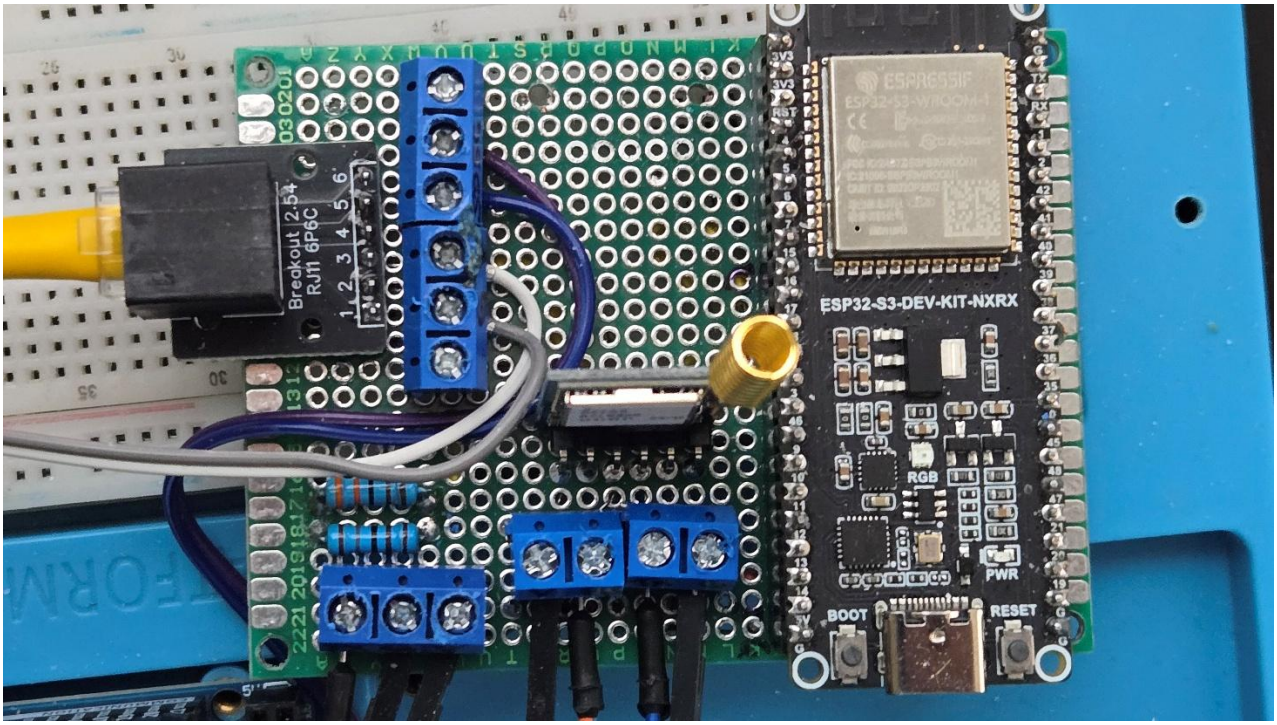

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**Figure 4-6. Main Station Controller Assembly**




*Main station electronics assembly showing Arduino Mega, LCD hat, yellow cable to sensors outside the office window, LoRa transceiver, and ESP32-S3 gateway interface board.*

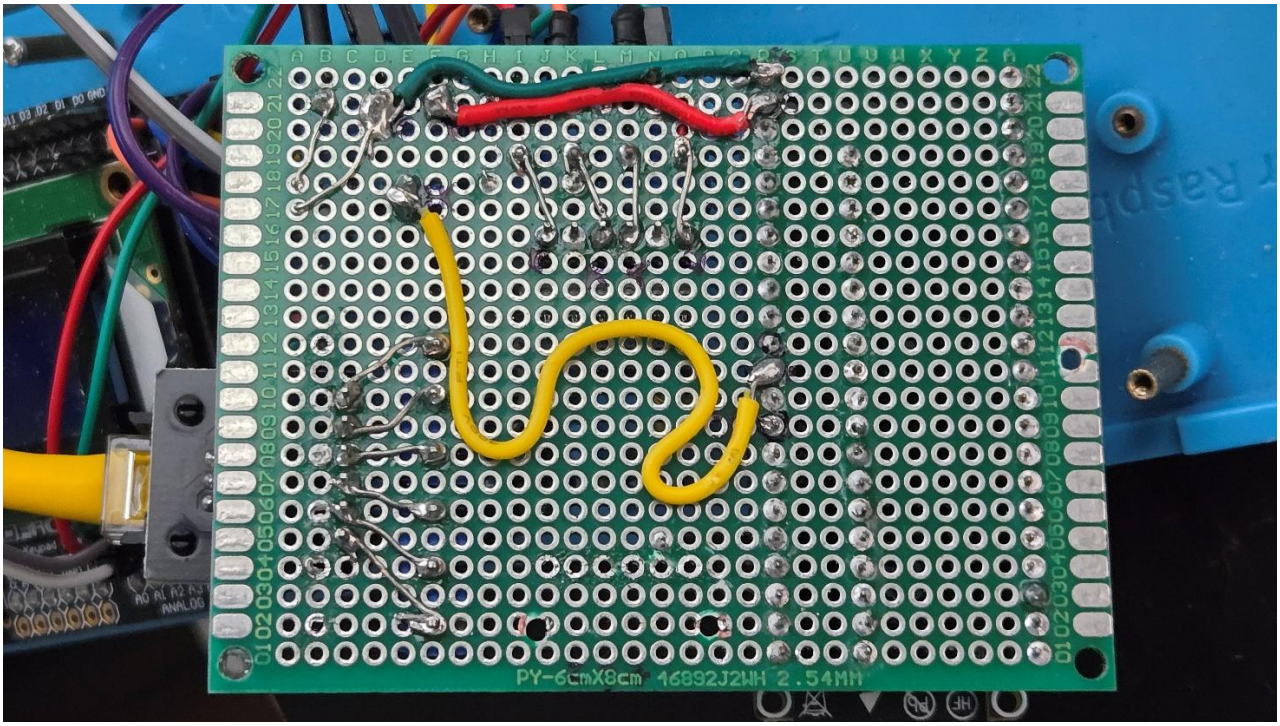
**Figure 4-7. Main Station Gateway Interface Board Top View**




*Top-side view of the ESP32-S3 gateway interface board showing the ESP32-S3 module, RYLR890 LoRa transceiver, RJ12 sensor interface, and supporting circuitry.*

*At lower left in this view, you can see the 1.8 kΩ / 3.3 kΩ resistor divider that is incorporated on the gateway interface board to reduce the Arduino Mega 5 V UART TX signal (D18) to an ESP32-S3 compatible 3.3 V level before connection to GPIO18 on the S3.*

**Figure 4-8. Main Station Gateway Interface Board Bottom View**




*Underside view showing point-to-point wiring used for power distribution and communications interfacing on the gateway board.*

*A 1.8 kΩ / 3.3 kΩ resistor divider reduces the Arduino Mega D18 (TX1) 5 V UART signal to an ESP32-S3 compatible 3.3 V level before connection to GPIO18.*

### 4.3 Mast-mounted Hardware Inventory

| Component | Function |
|---|---|
| ESP32-C3 | Controller |
| Wind Vane | Direction sensing |
| Anemometer | Wind-speed sensing |
| Rain Gauge | Rainfall measurement |
| RYLR890 | LoRa communications |
| Antenna | RF communications |

---

**Figure 4-9. Mast-mounted Electronics Diagram**

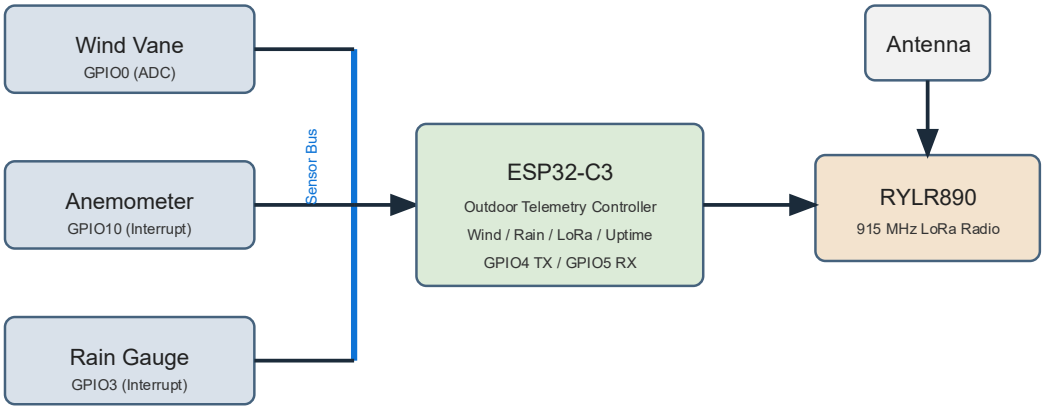


**Figure 4-10. Sensor Mast**



*Sensor mast showing weather sensors and supporting hardware.*

---

### 4.4 Arduino Mega GPIO Assignments

| Pin | Function |
|---|---|
| **A0** | **LCD keypad input** |
| **D10** | **LCD backlight** |
| **D16** | **LoRa TX** |
| **D17** | **LoRa RX** |
| **D18** | **UART1 TX to ESP32-S3** |
| **D19** | **UART1 RX from ESP32-S3** |
| **D20** | **SDA** |
| **D21** | **SCL** |

**UART Interface**

The Arduino Mega communicates with the ESP32-S3 gateway using the hardware serial port **Serial1** operating at 9600 baud. The interface uses Arduino Mega pin **D18 (TX1)** to transmit weather and status data to the ESP32-S3 and **D19 (RX1)** to receive data from the gateway. On the ESP32-S3, the corresponding UART1 pins are **GPIO18 (RX)** and **GPIO17 (TX)**. This dedicated serial link provides the communications path between the weather-processing controller and the gateway subsystem.

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### 4.5 ESP32-S3 GPIO Assignments

| GPIO | Function |
|---|---|
| GPIO18 | UART RX |
| GPIO17 | UART TX |

---

### 4.6 ESP32-C3 GPIO Assignments

| GPIO | Function |
|---|---|
| GPIO10 | Anemometer |
| GPIO3 | Rain Gauge |
| GPIO0 | Wind Vane ADC |
| GPIO4 | LoRa RX |
| GPIO5 | LoRa TX |

---

### 4.7 Sensor Wiring Summary

#### Wind Vane

| Connection | Signal |
|---|---|
| Pin 2 | Ground |
| Pin 3 | 3.3V |
| Pin 5 | ADC |

#### Anemometer

| Connection | Signal |
|---|---|
| Pin 3 | GPIO10 |
| Pin 4 | Ground |

#### Rain Gauge

| Connection | Signal |
|---|---|
| Pin 3 | GPIO3 |
| Pin 4 | Ground |

---

### Signal Conditioning Components

| Signal | ESP32-C3 Pin | Conditioning Circuit |
|---|---|---|
| Wind Vane | GPIO0 (ADC) | 12K pull-up resistor to 3.3V |
| Rain Gauge | GPIO3 | 10K pull-up resistor to 3.3V |
| Anemometer | GPIO10 | Direct digital input |

A 12K pull-up resistor was selected to optimize ADC separation between vane resistor-network states.

---

**Figure 4-11. Solar Power System**

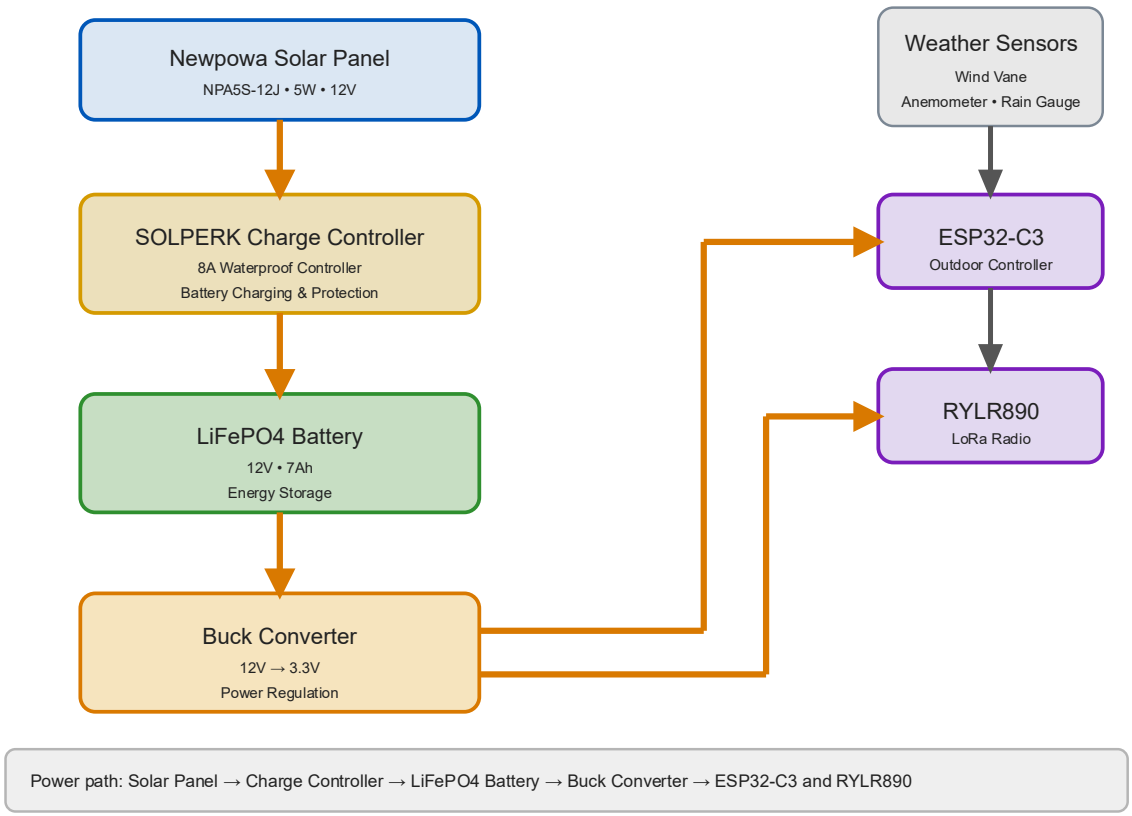


**Figure 4-12. Mast-mounted Solar Power Assembly**

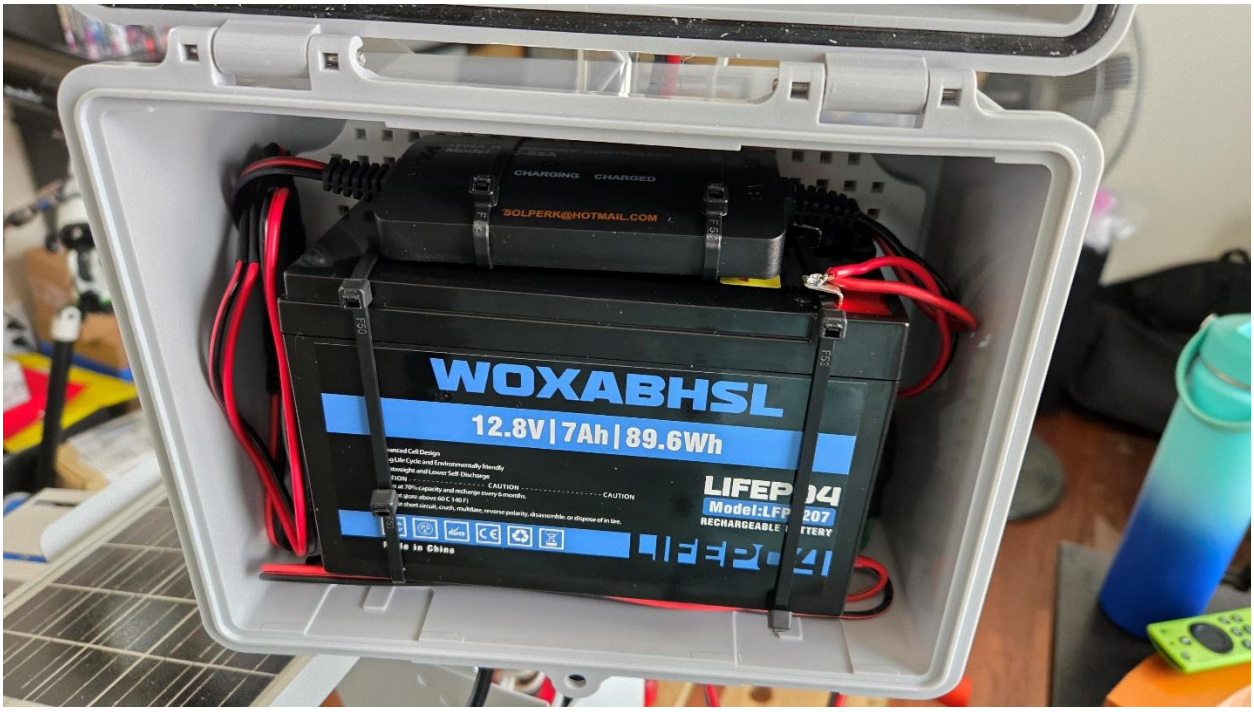
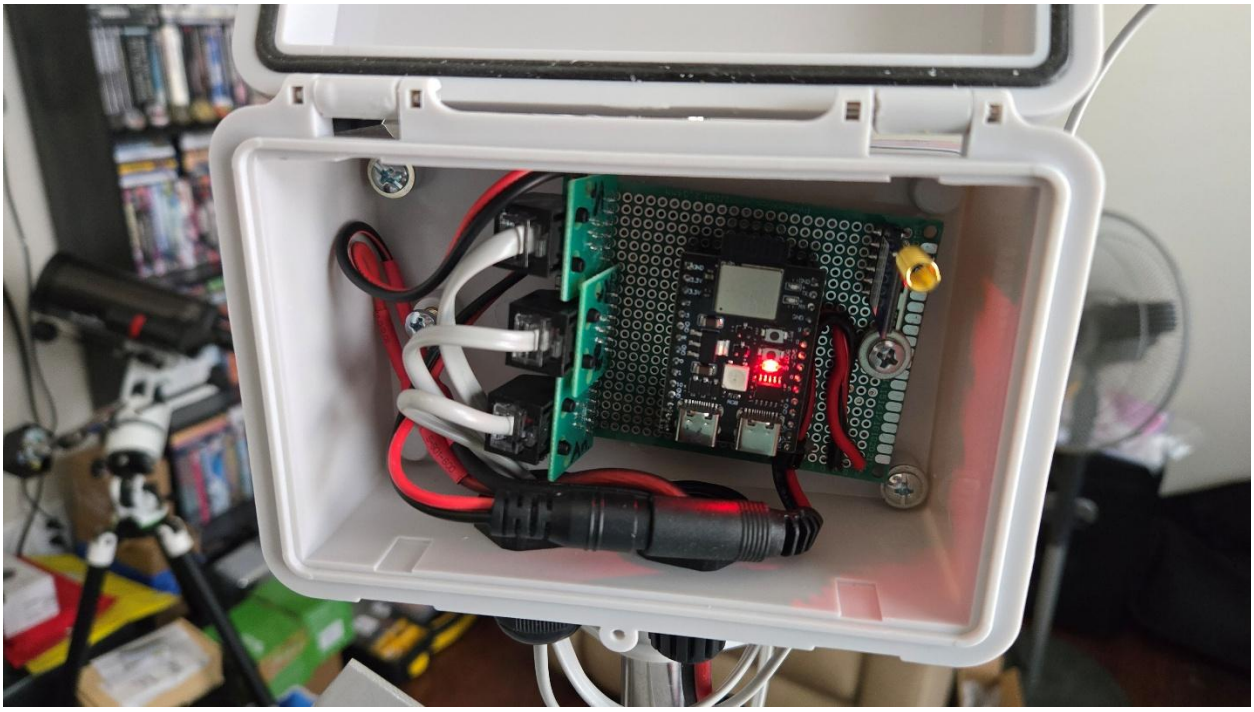


**Figure 4-13. Mast-mounted Controller Electronics Assembly**



These two photographs document the as-built implementation of the mast-mounted power and telemetry subsystems.

The solar power assembly contains the charge controller, LiFePO4 battery, wiring harnesses, and associated power-distribution components.

The controller electronics assembly contains sensor-interface wiring (RJ11 sockets), the ESP32-C3 controller, RYLR890 LoRa radio, buck converter, and signal-conditioning components.

These images provide a field-reference baseline for maintenance and troubleshooting.

(Configuration is shown as installed during Release 1.0 documentation.)

**Table 4-7 Solar Hardware Inventory**

| Component | Manufacturer | Model | Specification | Purpose |
|---|---|---|---|---|
| Solar Panel | Newpowa | NPA5S-12J | 5W, 12V Monocrystalline | Primary energy source |
| Charge Controller | SOLPERK | 8A Controller | IP67 Rated | Battery charging and protection |
| Battery | WOXABHSL | LFP1207 | 12V 7Ah LiFePO4 | Energy storage |
| Controller | ESP32-C3 | N/A | 3.3V Controller | Telemetry processing |
| LoRa Radio | Reyax | RYLR890 | 915 MHz | Wireless communications |
| Antenna | Compatible 915 MHz | N/A | LoRa antenna | Radio communications |

**Design Objectives**

The solar-power subsystem was selected to provide:

- Continuous sensor operation
- No external power wiring
- Flexible sensor placement
- Improved installation options
- Reduced maintenance requirements

The battery and charge controller are sized to maintain operation during periods of reduced solar availability while supporting continuous weather monitoring and LoRa communications.

---

**4.9 Communications Hardware**

The communications subsystem consists of:

- LoRa indoor module
- LoRa mast-mounted module
- LoRa antennas
- UART interface
- Wi-Fi interface

These components provide all data transport throughout the system.

---

**Sections 5–8**

**Firmware Architecture and Communications**

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## Section 5 – Arduino Mega Firmware Architecture

### 5.1 Purpose

The Arduino Mega serves as the central supervisory controller for the weather station. While the ESP32-C3 acquires weather measurements and the ESP32-S3 provides network and dashboard services, the Mega functions as the operational core of the system.

Responsibilities include:

- User-interface management
- Environmental sensor processing (from radiation shield and mast)
- RTC management
- EEPROM management
- Min/max record tracking
- Weather-data aggregation
- Communications coordination
- Fault-recovery management

The Mega is therefore the primary authority for weather-system state and operational control.

---

### 5.2 Software Architecture

The firmware utilizes a cooperative multitasking model built around the Arduino loop() function.

Rather than relying on a real-time operating system, the firmware schedules periodic activities through millisecond timing intervals using:

C++

millis()

This approach allows multiple independent activities to operate concurrently without blocking user interaction.

Major tasks include:

Sensor Acquisition

LCD Updates

Button Monitoring

LoRa Reception

ESP32 Packet Generation

EEPROM Management

Min/Max Tracking

Recovery Management

Diagnostics

---

**5.3 User Interface Architecture**

**LCD Hardware**

The user interface utilizes a standard LCD Keypad Shield mounted directly on the Arduino Mega.

The shield provides:

- 16x2 LCD display
- Five navigation buttons
- Backlight control

Buttons are connected through a resistor ladder network monitored on analog input A0.

---

**Button Functions**

| Button | Function |
|---|---|
| RIGHT | Navigate Forward |
| LEFT | Navigate Back |

| | |
|---|---|
| UP | Value Increase / Menu Navigation |
| DOWN | Value Decrease / Menu Navigation |
| SELECT | Accept / Enter |

---

**Menu Structure**

The firmware implements a hierarchical menu system.

Primary screens:

Super Screen

Wind Screen

Rain Screen

Clock Screen

Pressure Screen

Min Temperature

Max Temperature

Humidity Records

Pressure Records

Settings

Diagnostics

Button Test

The menu system provides complete access to operational and diagnostic functions without requiring network connectivity.

---

**5.4 Sensor Acquisition Engine**

**SHT31-D**

Primary measurements:

- Temperature
- Relative Humidity

The SHT31-D serves as the preferred environmental sensor because it provides both measurements from a single source.

---

**BMP280**

Measurements:

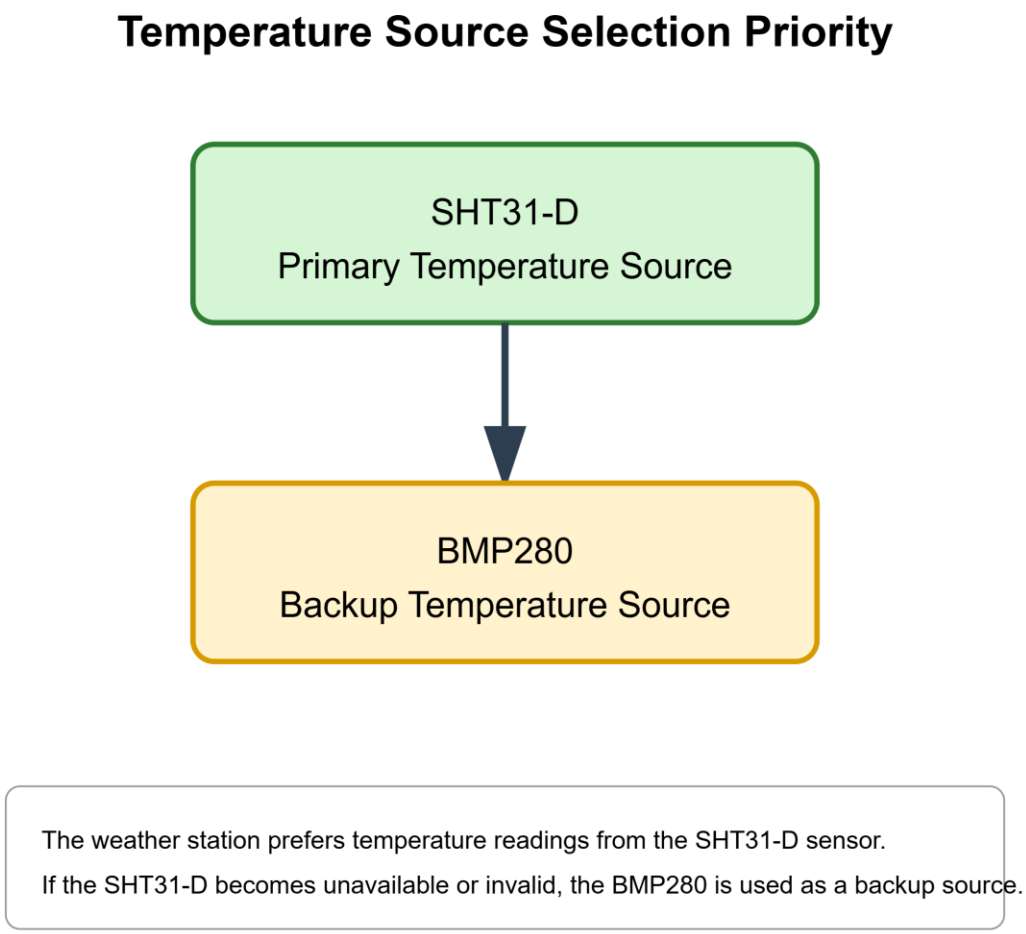
- Atmospheric Pressure
- Backup Temperature

The BMP280 primarily supplies pressure measurements but also provides secondary temperature data.

---

**Sensor Redundancy**

Temperature processing includes source-selection logic.




If the primary source becomes invalid, the firmware automatically transitions to the backup source.

This improves system resilience and reduces data loss.

---

**5.5 Min/Max Tracking Engine**

The weather station continuously tracks environmental extremes.

Tracked values:

Minimum Temperature

Maximum Temperature

Minimum Humidity

Maximum Humidity

Minimum Pressure

Maximum Pressure

Associated timestamps are stored for every record.

---

**Timestamp Components**

Records contain:

Month

Day

Hour

Minute

allowing operators to determine when the extreme occurred.

---

**5.6 Temperature Suppression System**

**Objective**

The suppression system compensates for known afternoon temperature influences that can elevate readings beyond representative ambient conditions.

---

**Operating Window**

The correction operates during:

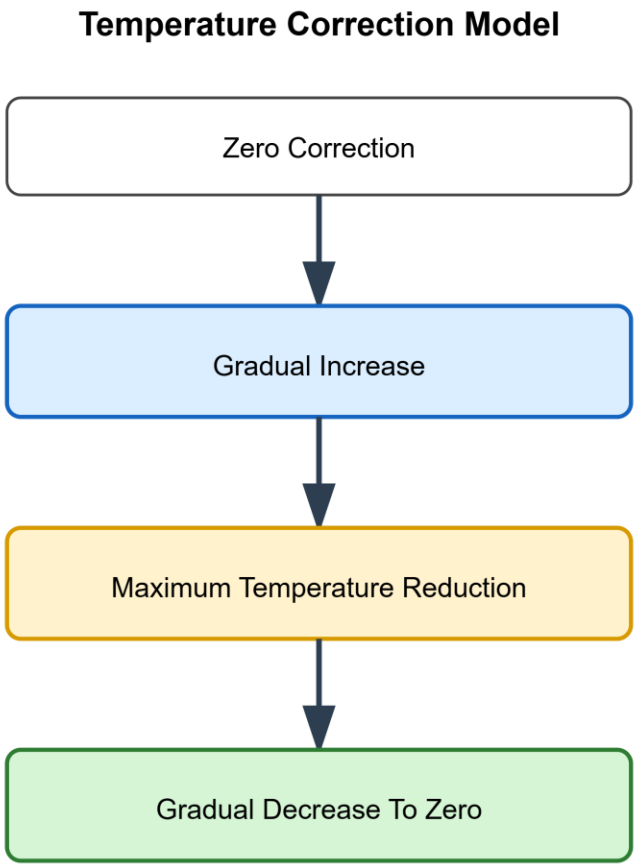
15:00 to 20:00

---

**Correction Model**

The adjustment follows a smooth sinusoidal curve.

Conceptually:




This avoids abrupt transitions while maintaining consistent environmental reporting.

---

**5.7 EEPROM Manager**

The firmware stores persistent data in EEPROM.

Stored information includes:

Min Temperature

Max Temperature

Min Humidity

Max Humidity

Min Pressure

Max Pressure

Altitude

Reset Information

EEPROM preservation ensures critical information survives power interruptions.

---

**5.8 RTC Management**

The DS3231 RTC provides:

- Date
- Time
- Daily rollover detection

The RTC acts as the authoritative time source for:

Weather Records

Database Uploads

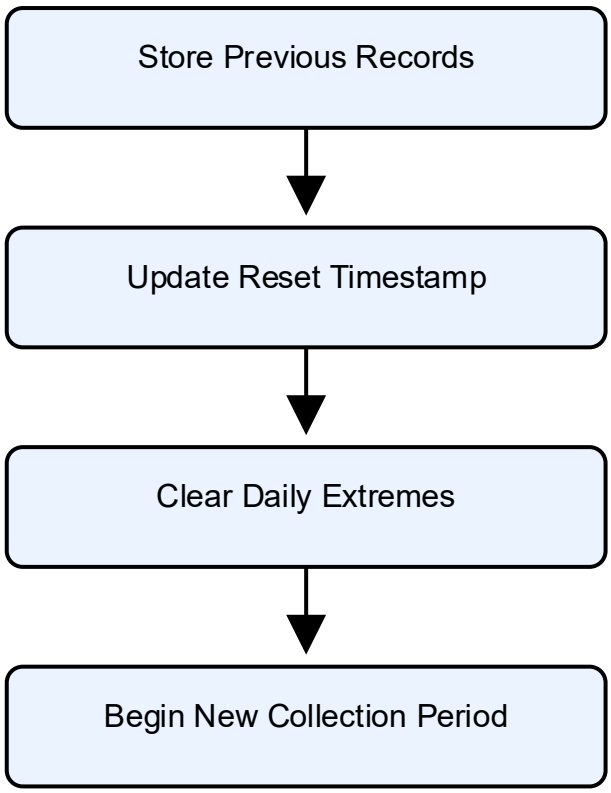
Min/Max Timestamps

Daily Resets

---

**5.9 Daily Reset Processing**

At day rollover:




This process ensures clean separation between daily observation periods.

---

**5.10 Mast-mounted Communications Supervision**

The Mega continuously evaluates mast-mounted system telemetry health.

Monitoring criteria include:

Packet Arrival

Packet Freshness

ESP32-C3 Uptime

Recovery State

**Warning Threshold**

30 Seconds

without valid updates.

**Recovery Threshold**

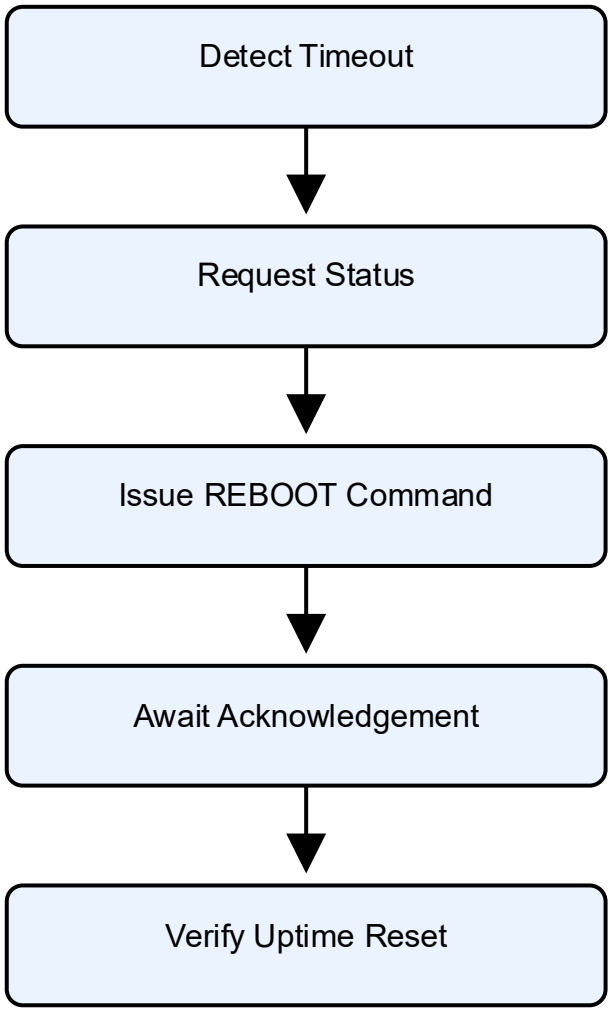
60 Seconds

without valid updates.

**5.11 Remote Reboot Manager**

If communications are lost, the Mega can initiate recovery.

Process:




Maximum retries: 3

---

### 5.12 Weather Packet Generation

Once all measurements are consolidated, the Mega generates a weather-report packet for the ESP32-S3.

Typical contents include:

Temperature

Humidity

Dew Point

Pressure

Wind Speed

Wind Gust

Wind Direction

Rainfall

Rain Rate

Daily Rain

ESP32-C3 Uptime

Date

Time

Transmission interval:  Approximately 5 seconds

---

## Section 6 – ESP32-S3 Gateway Firmware

### 6.1 Purpose

The ESP32-S3 acts as the Internet gateway and data-presentation platform.

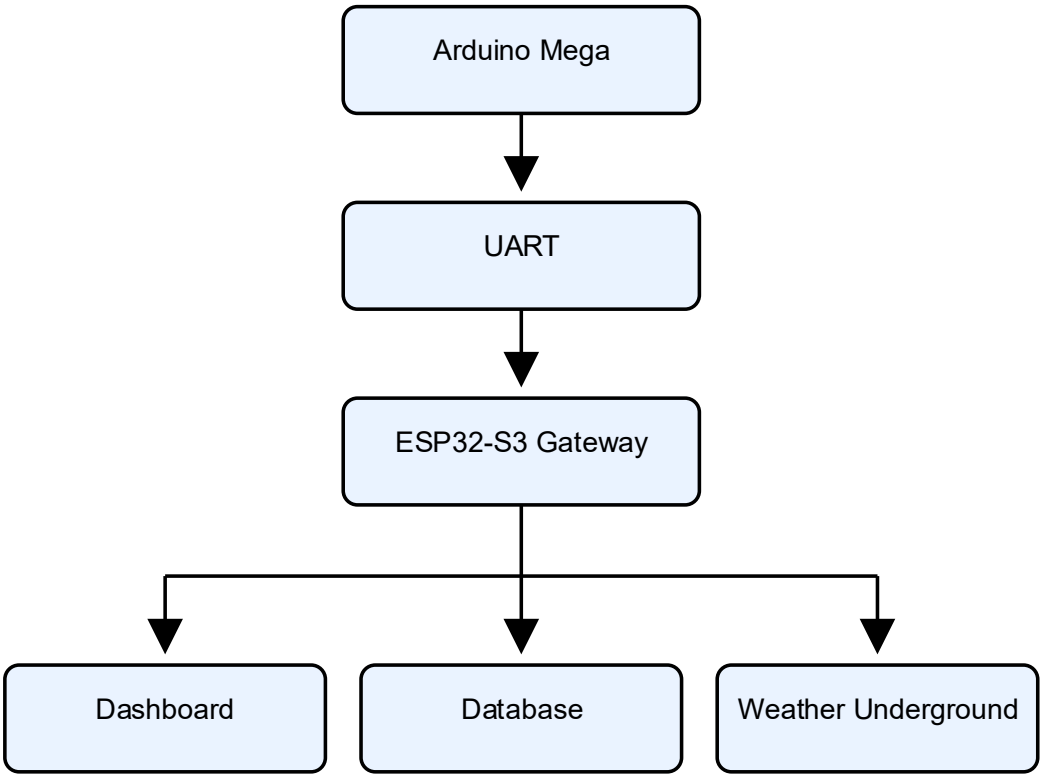
Responsibilities include:

- Wi-Fi connectivity

- Dashboard generation
- Database uploads
- Weather Underground uploads
- Status monitoring
- Communications supervision

---

**6.2 Gateway Data Flow**




The gateway does not generate weather data.

Instead, it receives processed information from the Mega.

---

**6.3 UART Processing Engine**

The ESP32-S3 continuously monitors UART traffic from the Mega.

The parser extracts:

Temperature

Humidity

Pressure

Wind Data

Rain Data

System Status

These values become dashboard variables.

---

**6.4 Dashboard Engine**

The gateway hosts a browser-accessible dashboard.

Displayed information includes:

Temperature

Humidity

Dew Point

Pressure

Wind

Rain

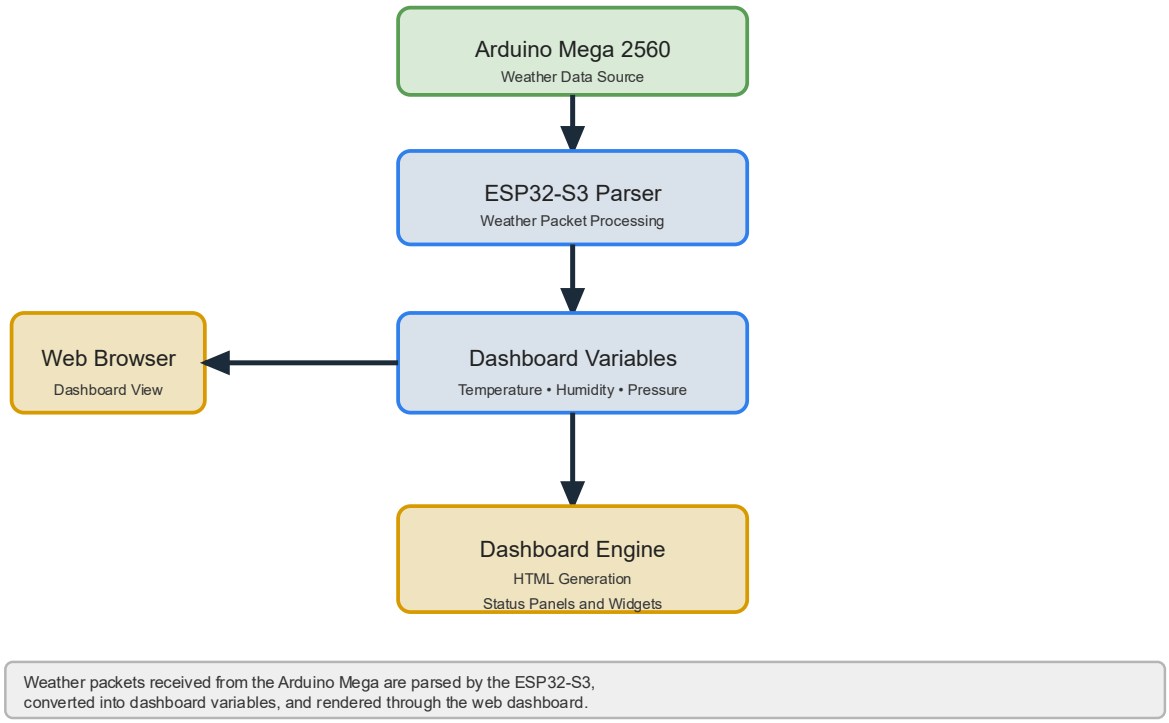
ESP32-C3 Uptime

Weather Underground Status

The dashboard serves as the primary remote-monitoring interface.

---

**Figure 6-1. Dashboard Data Flow**




Processing path used by the ESP32-S3 gateway to transform weather packets into dashboard content.

---

### 6.5 Weather Underground Integration

The ESP32-S3 performs all Weather Underground communications.

Functions:

Generate Upload Request

Transmit Data

Evaluate Success

Update Status

Dashboard classifications:

ONLINE

DEGRADED

OFFLINE

---

**6.6 Database Upload Engine**

The gateway uploads weather information to:

weatherdb.php

Data categories include:

- Environmental values
- Health metrics
- Upload status
- Time information

---

**6.7 Moisture Analysis**

The gateway calculates:

Dew Potential

Moisture Classification

These calculated values simplify interpretation of humidity data.

---

**6.8 Health Monitoring**

The gateway tracks:

Wi-Fi Status

Database Status

Weather Underground Status

Upload Success

These values provide early warning of communications issues.

---

**6.9 Recovery Logic**

If communications failures exceed acceptable limits, the ESP32-S3 may initiate self-recovery procedures.

Goals:

- Restore uploads
- Restore dashboard operation
- Restore connectivity

---

**Section 7 – ESP32-C3 Node Firmware**

**7.1 Purpose**

The ESP32-C3 functions as the mast-mounted telemetry controller.

Responsibilities include:

- Wind-speed measurement
- Wind-direction processing
- Rainfall measurement
- LoRa communications
- Uptime reporting

---

**7.2 Sensor Interfaces**

**Wind Vane**

GPIO0

ADC-based sensing.

---

**Anemometer**

GPIO10

Interrupt-driven pulse counting.

---

**Rain Gauge**

GPIO3

Interrupt-driven pulse counting.

---

**7.3 Wind Measurement System**

The anemometer generates reed-switch transitions.

Process:

Pulse Detection

↓

Count Accumulation

↓

Wind-Speed Calculation

---

**7.4 Gust Processing**

The firmware continuously tracks peak instantaneous wind activity.

Result:

Maximum Gust

reported within outgoing telemetry packets.

---

**7.5 Wind-Direction Processing**

The wind vane produces a variable resistance network.

The firmware converts:

ADC Reading

into

Compass Direction

in

Degrees

before transmission.

---

**7.6 Rainfall Processing**

The tipping-bucket rain gauge generates pulse events.

Measurements include:

Rain Tips

Rain Rate

Daily Rainfall

---

**7.7 LoRa Packet Construction**

The mast-mounted node builds telemetry packets approximately every five seconds.

Packet contents:

Wind Speed

Wind Gust

Wind Direction

Rain Rate

Daily Rain

Rain Tips

Uptime

---

**7.8 Uptime Monitoring**

The ESP32-C3 continuously reports uptime.

Benefits:

- Health monitoring
- Recovery verification
- Communications diagnostics

---

**7.9 Remote Command Processing**

Supported commands:

STATUS

REBOOT

These commands allow remote supervision by the Arduino Mega.

---

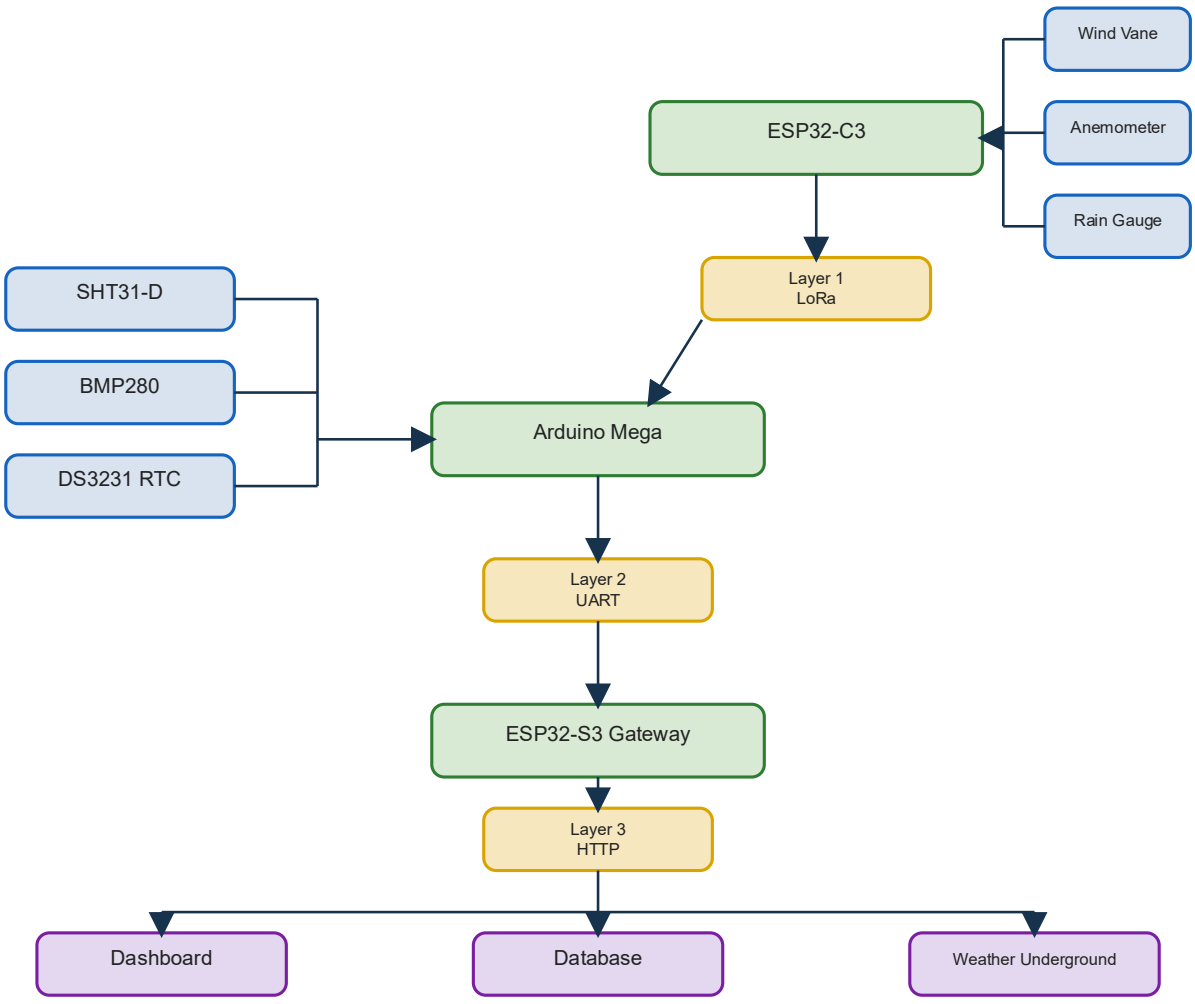
**Section 8 – Communications Protocols**

**8.1 Purpose**

The communications subsystem links all weather-station components into a unified architecture.

---

**Figure 8-1. Communications and Data Acquisition Architecture**




End-to-end weather-data flow from sensors through the control system and into dashboard and cloud services.

---

### 8.2 Communications Layers

**Layer 1**

LoRa

Mast-mounted telemetry.

---

**Layer 2**

UART

Mega-to-gateway communications.

---

**Layer 3**

HTTP

Internet communications.

---

**8.3 LoRa Weather Packet**

Packet identifier:

<OUT>

Structure:

WindSpeed,

WindGust,

Direction,

RainRate,

DailyRain,

RainTips,

Uptime

Update interval:  ~5 Seconds

---

**8.4 Status Protocol**

Query:

<STATUS?>

Response:

<STATUS>

Purpose:

Verify node health when telemetry becomes stale.

---

**8.5 Remote Reboot Protocol**

Command:

<REBOOT>

Acknowledgement:

<REBOOTING>

Verification:

Uptime Reset

---

**8.6 UART Weather Packet**

The Mega forwards consolidated weather information to the ESP32-S3.

Fields include:

Temperature

Humidity

Pressure

Wind

Rain

Uptime

Date

Time

---

**8.7 HTTP Uploads**

Destination services:

Database

Weather Underground

The ESP32-S3 performs all upload operations.

---

**8.8 Communications Timing**

| **Activity** | **Interval** |
|---|---|
| LoRa Telemetry | ~5 sec |
| UART Update | ~5 sec |
| Status Check | ~30 sec |
| Reboot Trigger | ~60 sec |

| | |
|---|---|
| Retry Count | 3 |

---

**8.9 Communications Health Monitoring**

The architecture continuously monitors:

LoRa Health

Gateway Status

Upload Status

Database Connectivity

Weather Underground Connectivity

This layered monitoring structure provides robust fault detection and recovery.

---

**Sections 9–14**

**Database, Dashboard, Algorithms, Installation, Troubleshooting, and Maintenance**

---

**Section 9 – Database Design**

**9.1 Purpose**

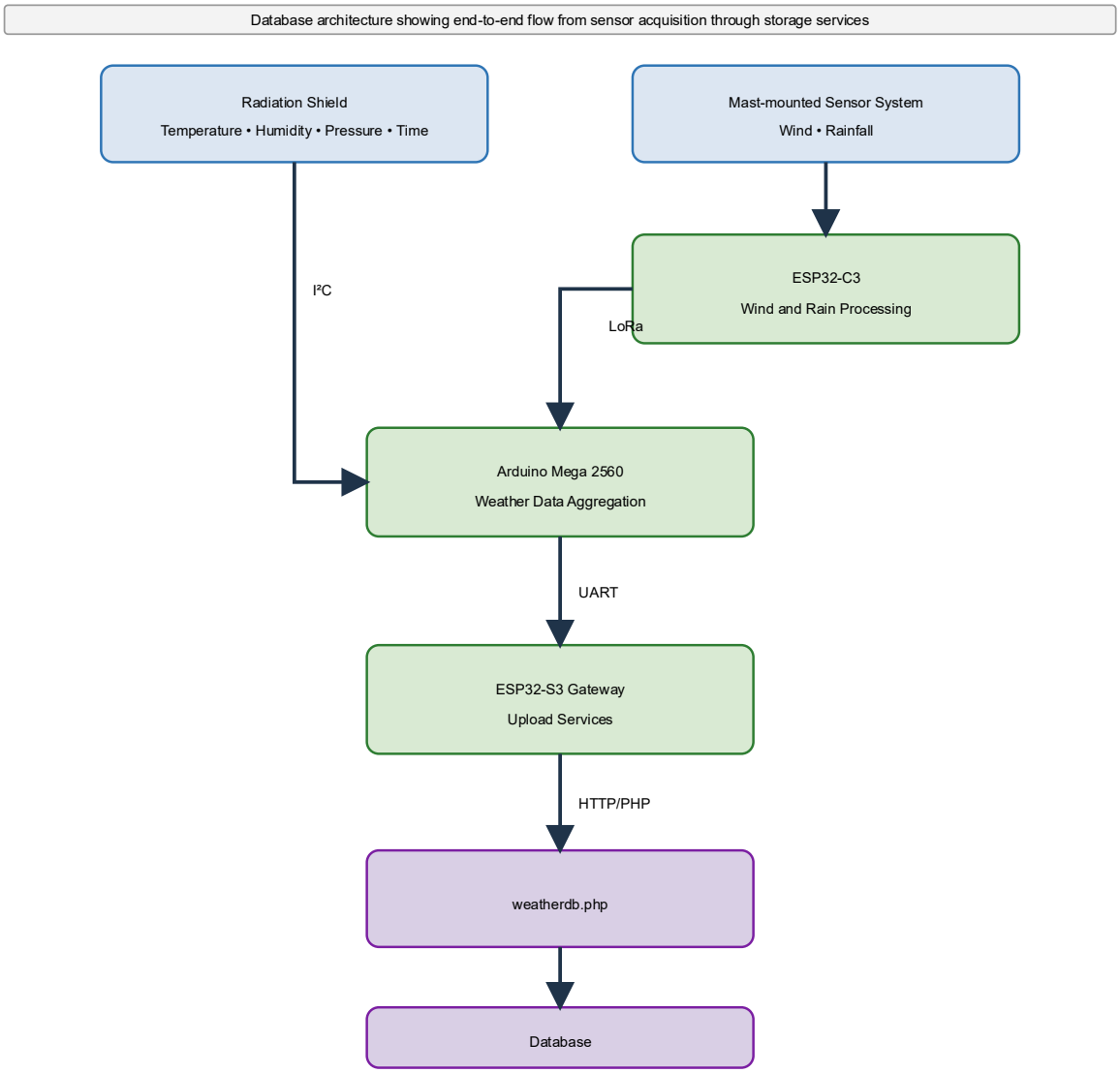
The database subsystem provides long-term storage of weather and operational data generated by the weather station. While the Arduino Mega and ESP32-S3 maintain current operating values, the database preserves historical observations for reporting, troubleshooting, trend analysis, and future dashboard enhancements.

The database serves as the permanent historical record of station operation.

---

### 9.2 Database Architecture

The database resides downstream of the weather-acquisition chain.




This separation allows weather acquisition to continue even when database services are unavailable.

---

### 9.3 Design Objectives

The database design was developed to meet several goals:

- Provide historical weather storage
- Preserve raw and corrected values

- Support future reporting tools
- Support dashboard displays
- Simplify troubleshooting
- Record system-health information

---

**9.4 Observation Records**

Each database record represents a single observation event.

Typical categories include:

Environmental Conditions

System Health

Upload Status

Timestamp Information

Each observation is timestamped and archived for later retrieval.

---

**9.5 Environmental Data**

Stored environmental information includes:

**Temperature**

temp

rawtemp

adjfactor

---

**Humidity**

humidity

dewpt

moist

dewpot

---

**Pressure**

presshpa

pressinhg

---

**Wind**

winddir

windspd

windgust

windlast

---

**Rain**

rainrate

rainday

raintips

---

**9.6 System-Health Data**

The database stores operational status information.

Examples:

ESP32-C3 Uptime

Weather Underground Status

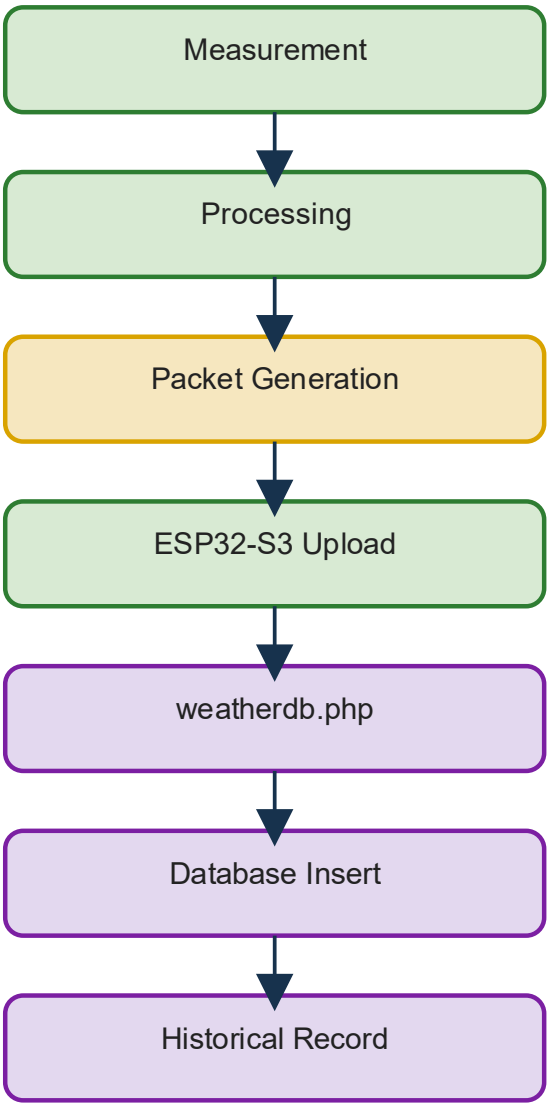
Last Connection

Next Upload Information

These values greatly simplify fault diagnosis.

---

### 9.7 Observation Lifecycle




---

### 9.8 Database Failure Tolerance

One design objective was fault isolation.

If database services fail:

Weather Collection Continues

LCD Continues

LoRa Continues

Dashboard Continues

Only historical record insertion is affected.

---

**9.9 Future Database Applications**

The current schema supports:

- Historical weather charts
- Rainfall reports
- Uptime analytics
- Reliability analysis
- Trend forecasting

without requiring major structural changes.

---

**Section 10 – Dashboard Design**

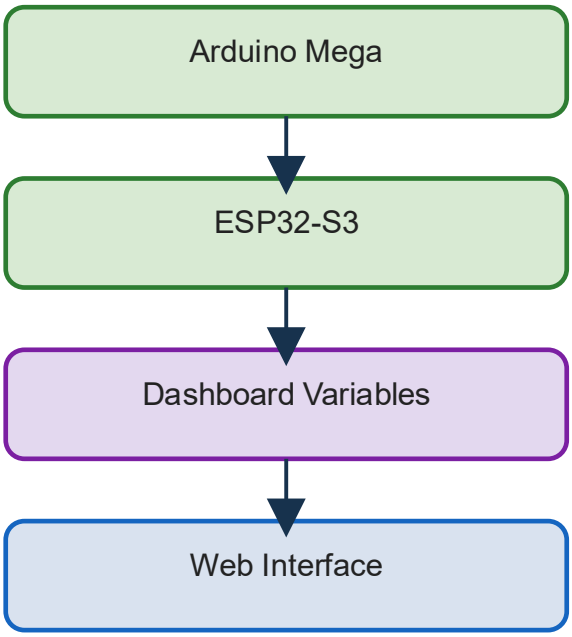
**10.1 Purpose**

The dashboard provides remote visibility into weather conditions and station health.

Unlike the LCD display, which is intended for local operation, the dashboard is accessible through a standard web browser and provides a complete system overview.

---

### 10.2 Dashboard Architecture




The dashboard reflects the most recent weather information received by the gateway.

---

### 10.3 Design Goals

The dashboard was designed to provide:

- Rapid readability
- Mobile compatibility
- System-health visibility
- Weather Underground visibility
- Near real-time updates

---

### 10.4 Main Dashboard Components

The dashboard displays:

Temperature

Humidity

Dew Point

Pressure

Wind

Rainfall

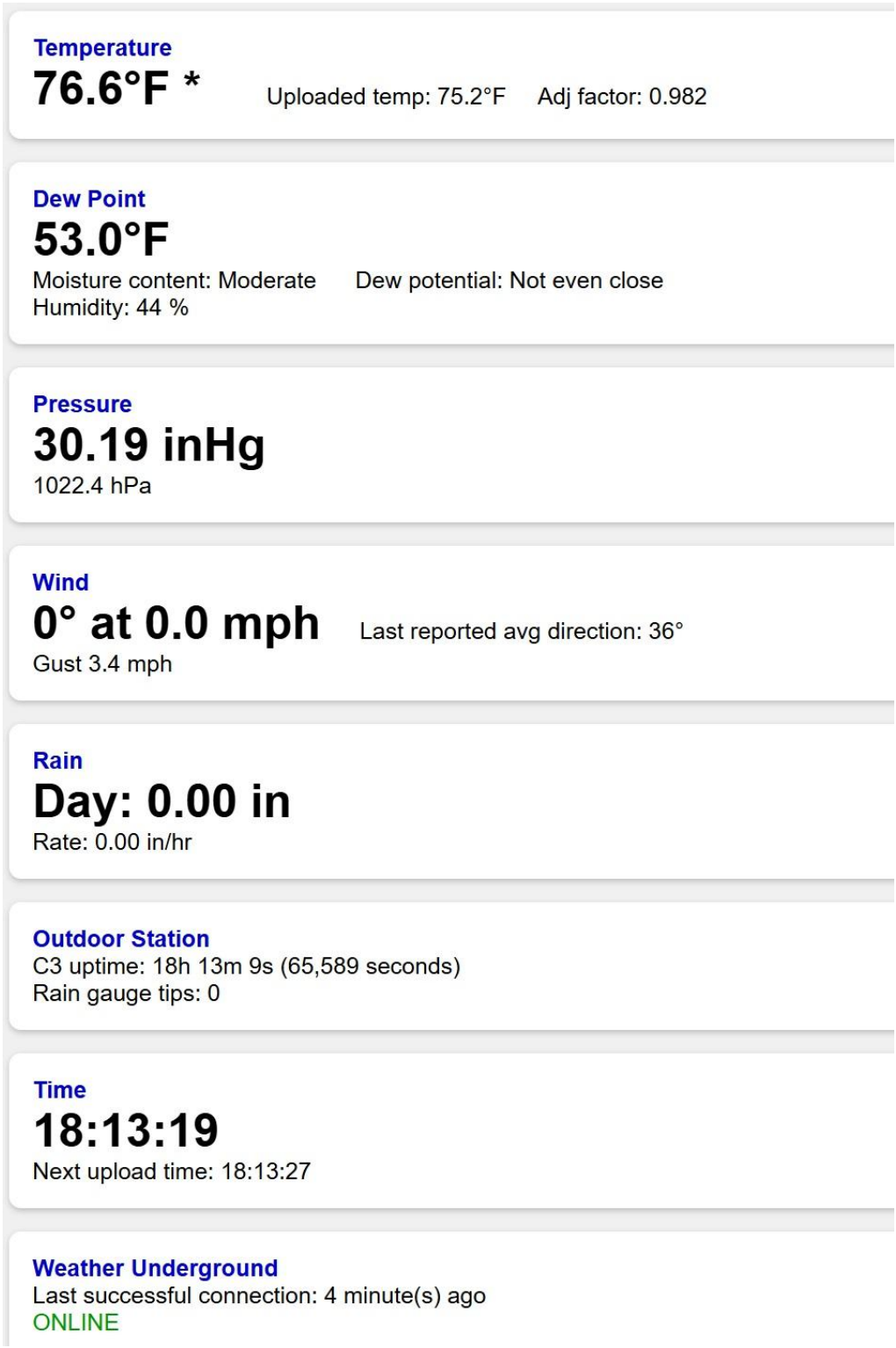
ESP32-C3 Uptime

Weather Underground Status

Date/Time

---

**Figure 10-1. Dashboard Screenshot**




*Primary localhost dashboard interface showing current weather and system status.*

---

**10.5 Temperature Display**

The temperature section shows:

- Corrected temperature
- Sensor-derived values
- Suppression-adjusted values

The displayed value represents the station's preferred weather temperature.

---

**10.6 Humidity and Dew Point**

The humidity section displays:

Relative Humidity

Dew Point

Moisture Classification

providing a quick understanding of atmospheric conditions.

---

**10.7 Pressure Display**

Pressure information is displayed using corrected sea-level pressure.

Available units include:

hPa

inHg

---

**10.8 Wind Display**

Wind information includes:

Current Speed

Wind Gust

Direction

These values originate from the mast-mounted node.

---

**10.9 Rainfall Display**

The rainfall section provides:

Rain Rate

Daily Rainfall

Rain-Gauge Status

This allows rapid assessment of precipitation conditions.

---

**10.10 Mast-mounted Node Status**

The dashboard displays:

ESP32-C3 Uptime

Communication Health

A rapidly growing uptime indicates healthy operation.

---

**10.11 Weather Underground Status**

Status indicators:

ONLINE

DEGRADED

OFFLINE

provide visibility into cloud-upload performance.

---

### 10.12 Future Dashboard Enhancements

Potential upgrades include:

- Historical charts
- Rainfall graphs
- Pressure trends
- Wind roses
- Battery telemetry

---

## Section 11 – Environmental Algorithms

### 11.1 Purpose

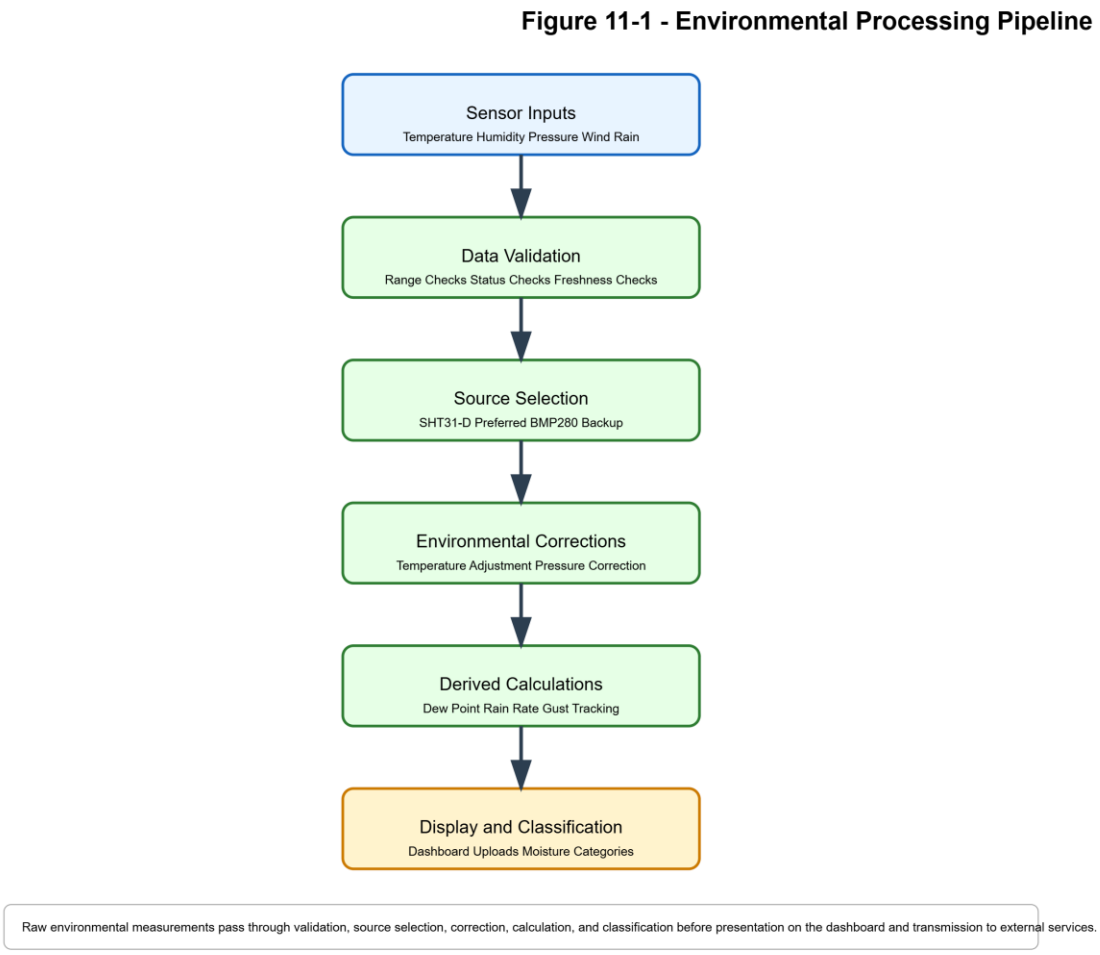
Environmental algorithms convert raw sensor measurements into useful meteorological information.

These algorithms improve usability while preserving scientific value.

---

### 11.2 Processing Pipeline

**Figure 11-1. Environmental Processing Pipeline**




Environmental-data processing sequence from raw sensor measurements through correction, calculation, classification, display, and upload.

---

### 11.3 Temperature Source Selection

Temperature may originate from:

SHT31-D

BMP280

The system automatically selects the preferred valid source.

---

### 11.4 Temperature Suppression Algorithm

The firmware contains a time-based correction system intended to minimize elevated readings during a known afternoon interval.

**Operating Window**

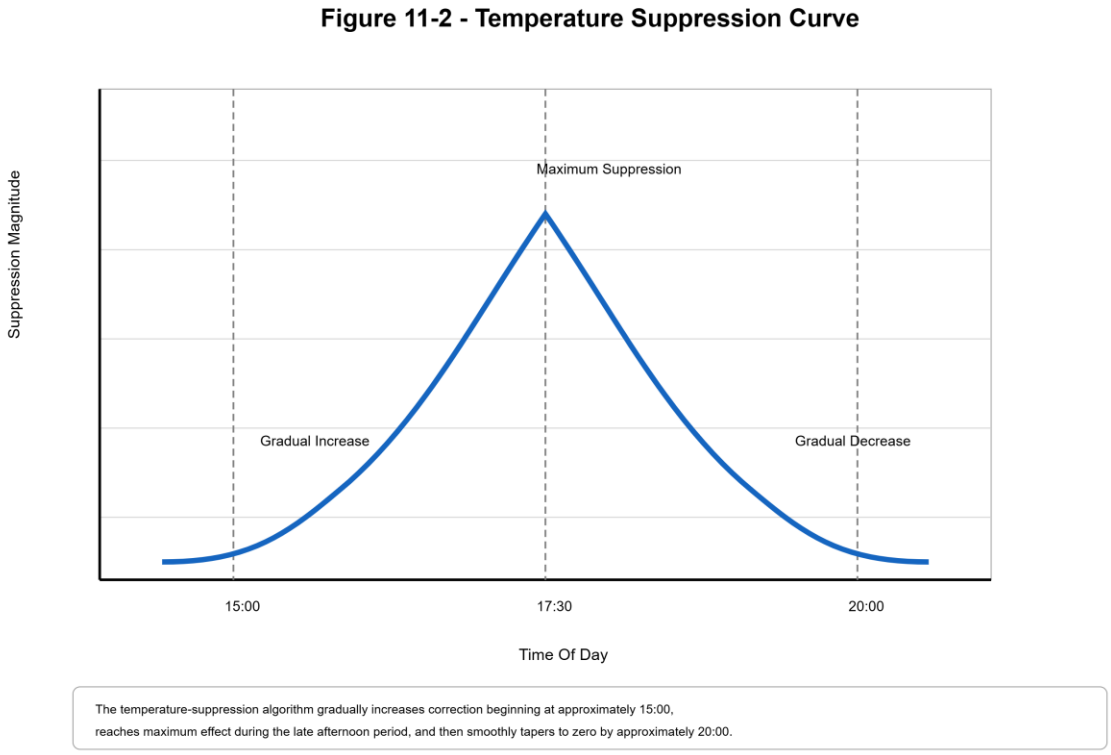
15:00 – 20:00

**Characteristics**

- Smooth transition
- Gradual increase
- Gradual decrease
- No abrupt step changes

---

**Figure 11-2. Temperature Suppression Curve**




Representative temperature-adjustment profile used to compensate for afternoon heating effects while avoiding abrupt transitions.

---

### 11.5 Dew-Point Calculation

Dew point is calculated using temperature and humidity.

General process:

Temperature

+

Humidity

↓

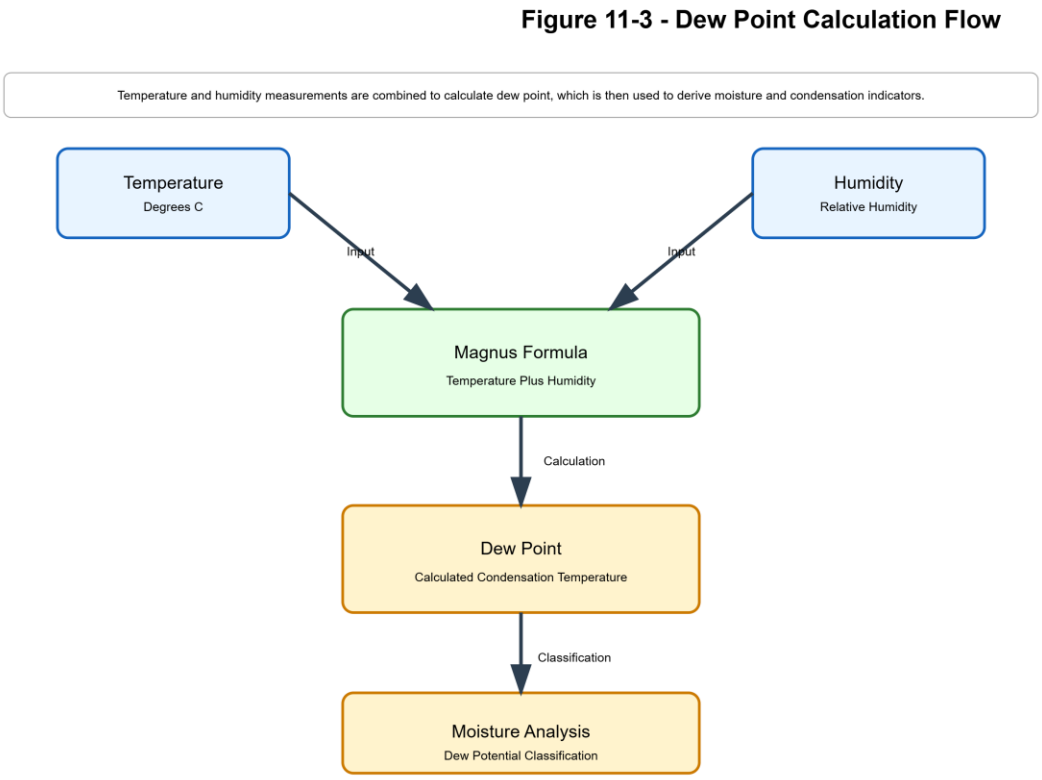
Magnus Equation

↓

Dew Point

---

**Figure 11-3. Dew Point Calculation Flow**




Dew-point generation process utilizing temperature and humidity measurements.

---

### 11.6 Pressure Correction

Pressure readings are corrected using station altitude.

**Inputs:**

Observed Pressure

Altitude

**Output:**

Sea-Level Pressure

This produces weather values comparable to nearby reporting stations.

---

**11.7 Wind Processing**

The mast-mounted node performs:

Pulse Counting

Wind-Speed Calculation

Gust Tracking

Direction Processing

before transmission.

---

**11.8 Rainfall Processing**

Rainfall calculations are derived from tipping-bucket events.

Generated values:

Rain Tips

Rain Rate

Daily Rainfall

---

**11.9 Moisture Classification**

The system generates human-readable environmental classifications from weather conditions.

Examples may include:

Dry

Comfortable

Humid

Very Humid

## Section 12 – Installation and Physical Deployment

### 12.1 Purpose

This section describes physical deployment of the complete weather station.

The installation process includes:

- Mechanical mounting
- Electronics installation
- Power-system installation
- Communications verification

---

### 12.2 Physical Assemblies

The station consists of:

Main Station

Radiation Shield Sensor Package

Mast-mounted Sensor System

Solar Power System

---

### 12.3 Site Selection

Ideal installation locations provide:

- Good wind exposure
- Open rainfall collection
- Reliable LoRa communications
- Sufficient solar exposure

---

**Figure 12-1. Sensor Mast Installation**



### 12.4 Wind Sensor Placement

Wind sensors should:

- Avoid obstructions
- Remain well above nearby structures
- Maintain unobstructed airflow

---

### 12.5 Rain Gauge Placement

The rain gauge should:

- Remain level
- Avoid roof runoff
- Avoid splash contamination
- Remain free of overhead obstructions

---

### 12.6 Radiation Shield Installation

The shield should:

- Remain ventilated
- Remain clean
- Avoid direct solar loading

---

**Figure 12-2. Radiation Shield**



### 12.7 Solar Power Installation

Install the solar power subsystem in accordance with Figures 4-11 through 4-13. Verify correct polarity between the solar panel, charge controller, battery, and buck converter before applying power.

---

### 12.8 Indoor Controller Installation

The indoor station should be installed:

- Indoors
- Within LoRa range
- Near power
- Accessible for maintenance

---

### 12.9 Commissioning Tests

Verify:

LoRa Communication

Dashboard Operation

Database Uploads

Weather Underground Uploads

Sensor Function

RTC Accuracy

before placing the station into service.

---

## Section 13 – Troubleshooting and Diagnostics

### 13.1 Purpose

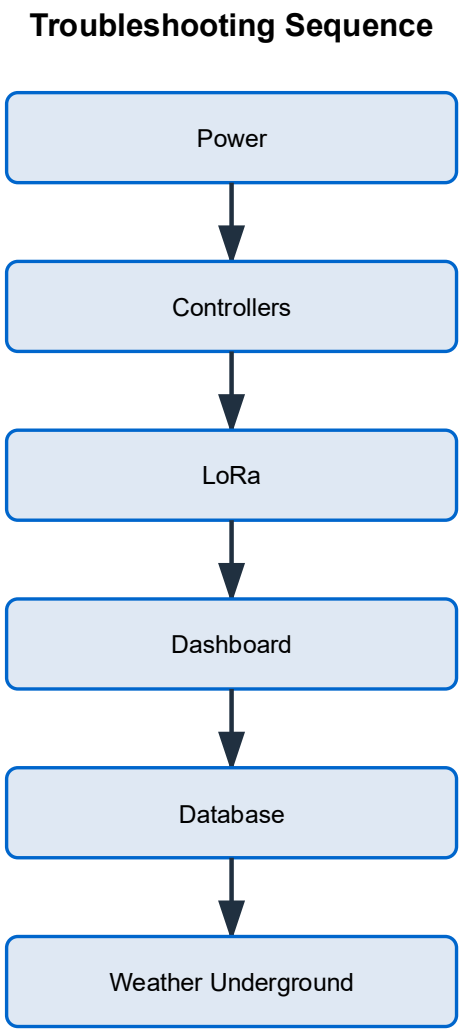
The troubleshooting process is designed to rapidly isolate system failures.

The architecture intentionally exposes diagnostic information through:

- LCD displays
- Dashboard indicators
- Uptime reporting
- Communications status

---

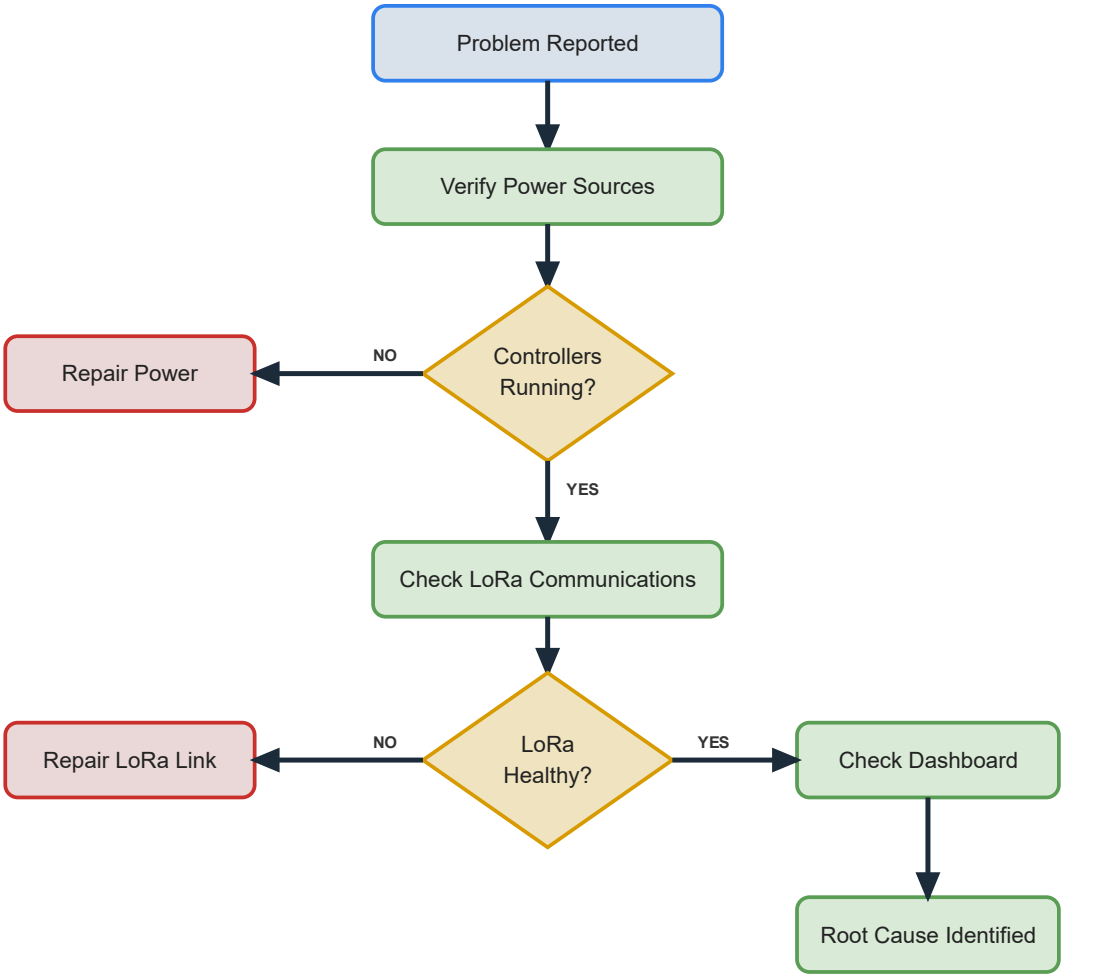
### 13.2 Troubleshooting Sequence




Recommended sequence

When diagnosing faults, begin with power verification and proceed sequentially through controllers, LoRa communications, dashboard operation, database services, and Weather Underground connectivity.

**Figure 13-1. Troubleshooting Workflow**




Follow the troubleshooting sequence from power verification through communications and dashboard validation to rapidly isolate the source of system failures.

Recommended fault-isolation process for weather-station troubleshooting.

---

### 13.3 MastNO Condition

Indicates stale mast-mounted communications.

Possible causes:

- Node outage
- LoRa failure
- Antenna issue
- Power-system failure

---

**13.4 Wind Data Missing**

Potential causes:

Sensor Failure

LoRa Failure

GPIO Wiring Problem

Inspection should begin with communications verification.

---

**13.5 Rainfall Problems**

Possible causes:

Debris

Bucket Binding

Wiring Failure

Pull-Up Resistor Failure

---

**13.6 Incorrect Time**

Potential causes:

RTC Configuration

RTC Battery

RTC Failure

---

**13.7 Dashboard Not Updating**

Possible causes:

UART Failure

Gateway Failure

Parser Failure

---

**13.8 Weather Underground Offline**

Potential causes:

Internet Outage

Wi-Fi Failure

DNS Failure

Credential Issue

---

**13.9 Database Upload Failure**

If dashboard updates continue but database records stop:

Likely Server-Side Failure

rather than weather acquisition failure.

---

**13.10 Failure-Isolation Matrix**

| Symptom | Likely Cause |
|---|---|
| MastNO | Mast-mounted Communication Failure |
| No Wind | Sensor / LoRa Problem |
| Frozen Dashboard | Gateway Problem |
| OFFLINE Status | Internet Problem |
| Missing DB Records | Server Problem |
| Repeated Reboots | Power Problem |

---

**Section 14 – Preventive Maintenance**

**14.1 Purpose**

Preventive maintenance maximizes system reliability and minimizes downtime.

Focus areas include:

Inspection

Cleaning

Verification

Testing

Documentation

---

**14.2 Recommended Maintenance Schedule**

| Activity | Interval |
|---|---|
| Dashboard Review | Weekly |
| Sensor Inspection | Monthly |

| | |
|---|---|
| Solar Cleaning | Monthly |
| Battery Inspection | Semiannual |
| Full Functional Test | Annual |

---

**14.3 Weekly Tasks**

Verify:

Dashboard Updating

ESP32-C3 Uptime Increasing

Weather Underground ONLINE

---

**14.4 Monthly Tasks**

Inspect:

Mast

Enclosure

Solar Panel

Mounting Hardware

Clean the solar panel as required.

---

**14.5 Quarterly Tasks**

Inspect:

Wind Vane

Anemometer

Rain Gauge

Sensor Wiring

Verify proper operation of each component.

---

**14.6 Semiannual Tasks**

Inspect:

Battery

Charge Controller

Power Wiring

Verify all connections remain secure.

---

**14.7 Annual Tasks**

Perform a complete functional verification.

Verify:

Sensors

LoRa

Dashboard

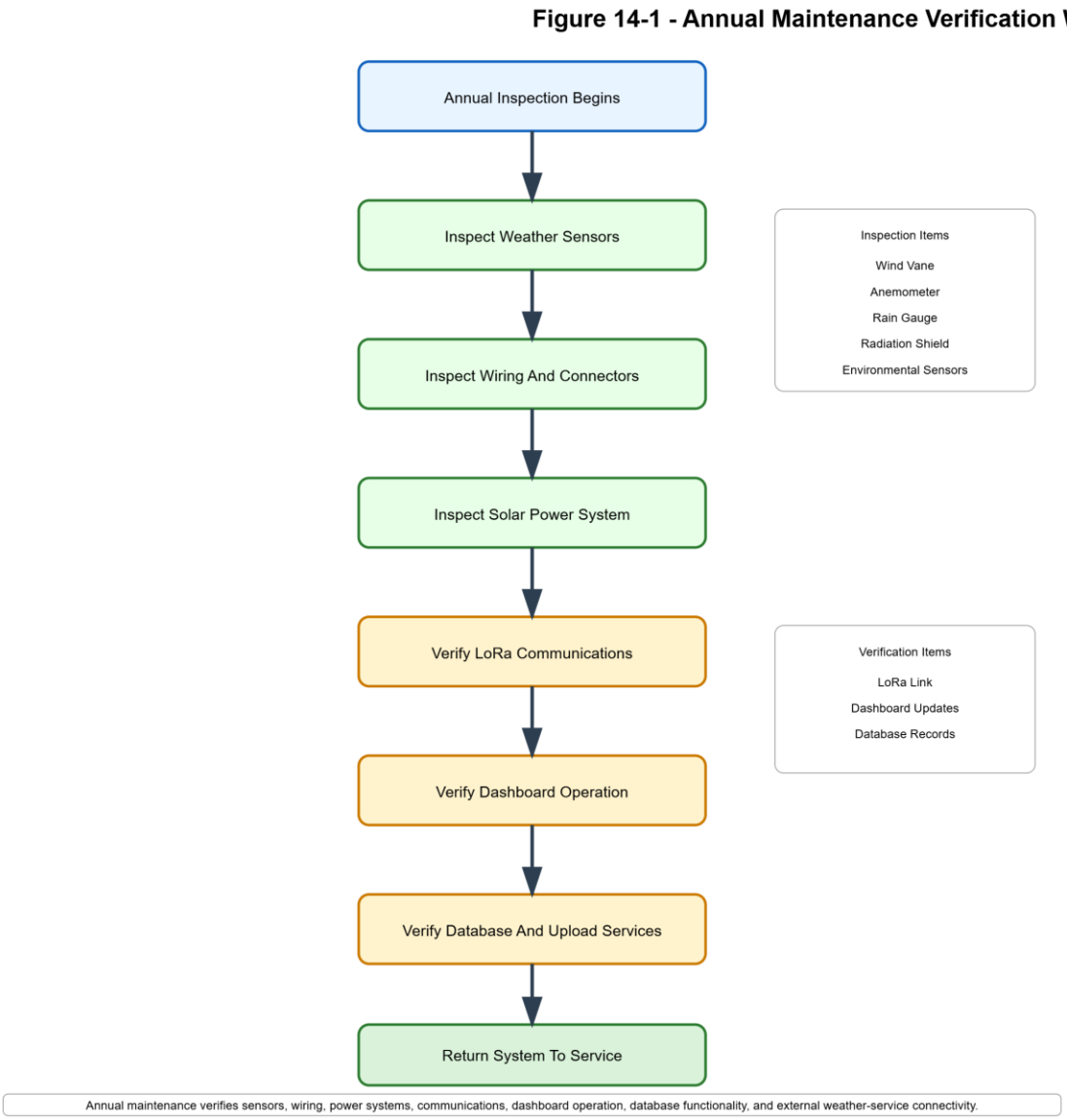
Database

Weather Underground

all remain operational.

---

**Figure 14-1. Annual Maintenance Verification Flow**




Annual maintenance verification sequence used to validate operation of all major system components.

---

### 14.8 Post-Storm Inspection

After severe weather:

Inspect:

Sensors

Pole

Enclosure

Solar Panel

Antenna

Cabling

and verify normal operation.

---

**14.9 Service Log**

A maintenance record should include:

Date

Technician

Findings

Corrective Actions

Firmware Version

Long-term records simplify support and system evolution.

---

**Section 15 – Operations and Long-Term Support**

**15.1 Purpose**

This section provides operational guidance that supplements the technical material presented throughout this manual. While previous sections focus on system architecture, firmware, communications, and hardware implementation, this section focuses on day-to-day operation, recovery procedures, preventive stewardship, and long-term maintenance.

The objective is to ensure that future maintenance personnel can successfully operate, troubleshoot, and preserve the Wireless Weather Station system even if the original designer is no longer available.

## 15.2 System Startup Procedure

### Cold Start

After a complete power loss, the system should start in the following sequence:

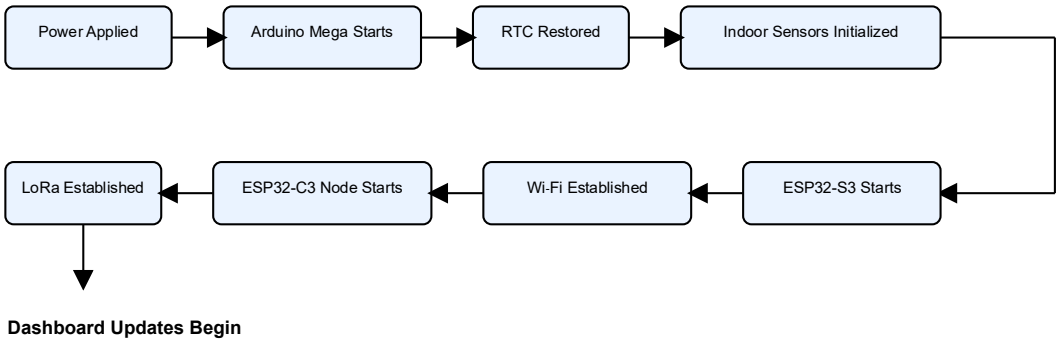



**Figure 15-1. Cold Start Initialization Sequence**

Normal startup should require no operator intervention.

### Startup Verification Checklist

Verify:

- LCD active
- Date and time valid
- MastOK displayed
- ESP32-C3 Uptime increasing
- Dashboard updating
- Weather Underground ONLINE
- Database records appearing

---

## 15.3 Normal Operations Checklist

The station is intended to operate continuously.

**Arduino Mega**

Verify:

- Display active
- Current weather displayed
- Button navigation functional

**ESP32-C3 Mast-mounted Node**

Verify:

- Uptime increasing
- Wind updates present
- Rain updates present

**ESP32-S3 Gateway**

Verify:

- Dashboard updating
- Database uploads active
- Weather Underground ONLINE

---

**15.4 Recovery Procedures**

**Communications Failure Recovery**

If mast-mounted communications stop:

1. Observe MastNO condition.
2. Verify ESP32-C3 uptime.
3. Allow automatic recovery.
4. Observe STATUS requests.
5. Observe REBOOT attempts.
6. Verify uptime reset after recovery.

**Dashboard Recovery**

If dashboard updates stop:

1. Verify Arduino Mega operation.
2. Verify UART communications.
3. Verify ESP32-S3 operation.
4. Verify browser connectivity.
5. Restart ESP32-S3 if necessary.

**Internet Recovery**

If uploads fail:

1. Verify Wi-Fi connectivity.
2. Verify local Internet access.
3. Verify Weather Underground status.
4. Verify database connectivity.
5. Observe automatic recovery behavior.

---

**15.5 Seasonal Inspection Recommendations**

**Spring**

- Rain-gauge cleaning
- Wiring inspection
- Sensor inspection

Particular attention should be given to winter-weather damage.

**Summer**

- Solar-panel cleaning
- Radiation-shield inspection
- Temperature verification

Excessive contamination can affect temperature accuracy.

**Fall**

- Rain-gauge cleaning
- Leaf removal
- Sensor inspection

This period is particularly important for maintaining rainfall accuracy.

**Winter**

- Battery inspection
- Solar charging verification
- Enclosure-seal inspection

Cold-weather operation should be verified before severe weather arrives.

---

**15.6 Documentation and Backup Strategy**

The following records should be preserved throughout the life of the system:

**Technical Documents**

- Technical Manual
- User Manual
- Protocol Specifications
- Wiring Diagrams

**Firmware Archives**

- Arduino Mega firmware
- ESP32-S3 firmware
- ESP32-C3 firmware

All released versions should be archived.

**Installation Records**

- Installation photographs
- Wiring photographs
- Enclosure photographs

- Sensor alignment notes

These records significantly simplify future troubleshooting and upgrades.

**Recommended Backup Schedule**

| Item | Recommended Frequency |
|---|---|
| Firmware | Before major changes |
| Technical Documentation | After each revision |
| Database | Weekly |
| Configuration Files | After modification |

Historical weather data becomes increasingly valuable as the archive grows.

---

**15.7 Spare Equipment Strategy**

Recommended on-hand inventory:

| Quantity | Item |
|---|---|
| 1 | Arduino Mega |
| 1 | ESP32-S3 |
| 1 | ESP32-C3 |
| 1 | LoRa Module |
| 10 | RJ11/RJ12 Connectors |
| 2 | Sensor Cables |
| 1 | LiFePO4 Battery |
| Assorted | Resistors and wiring components |

The objective is to enable rapid recovery from common hardware failures.

---

### 15.8 Long-Term Support Philosophy

The weather station was designed around three primary goals:

- Reliability
- Serviceability
- Expandability

Future modifications should preserve these principles whenever practical.

New features should not compromise:

- Existing weather acquisition
- Existing communications
- Existing recovery mechanisms

The preferred approach is evolutionary improvement rather than architecture replacement.

---

## Appendices A–H

## Reference Material, Asset Inventory, Change History, and Future Planning

---

## Appendix A – GPIO and Signal Reference

### A.1 Purpose

This appendix provides a consolidated reference for all controller GPIO assignments, communication interfaces, sensor connections, and signal paths used throughout the weather station.

The GPIO reference serves as the primary hardware reference for:

- Troubleshooting
- Firmware development
- Hardware expansion
- Maintenance
- System replication

---

### A.2 Arduino Mega GPIO Assignments

#### User Interface

| Pin | Function |
|---|---|
| A0 | LCD button ladder input |
| D10 | LCD backlight control |

#### LoRa Communications (Serial2)

| Pin | Function |
|---|---|
| D16 | LoRa TX to RYLR890 |
| D17 | LoRa RX from RYLR890 |

#### ESP32-S3 Communications (Serial1)

| Pin | Function |
|---|---|
| D18 | UART1 TX to ESP32-S3 |
| D19 | UART1 RX from ESP32-S3 |

#### $I^2C$ Bus

| Pin | Function |
|---|---|
| D20 | SDA |
| D21 | SCL |

---

### A.3 ESP32-S3 GPIO Assignments

| GPIO | Function |
|---|---|
| GPIO18 | UART1 RX from Arduino Mega D18 |
| GPIO17 | UART1 TX to Arduino Mega D19 |

**Serial Interface: UART1, 9600 baud.**

---

### A.4 ESP32-C3 GPIO Assignments

| GPIO | Function |
|---|---|
| GPIO0 | Wind vane ADC input |
| GPIO3 | Rain gauge input |
| GPIO4 | LoRa TX to RYLR890 RX |
| GPIO5 | LoRa RX from RYLR890 TX |
| GPIO10 | Anemometer input |

---

### A.5 Sensor Interface Summary

**Radiation Shield Sensors**

| Device | Interface |
|---|---|
| SHT31-D | $I^2C$ |
| BMP280 | $I^2C$ |
| DS3231 RTC | $I^2C$ |

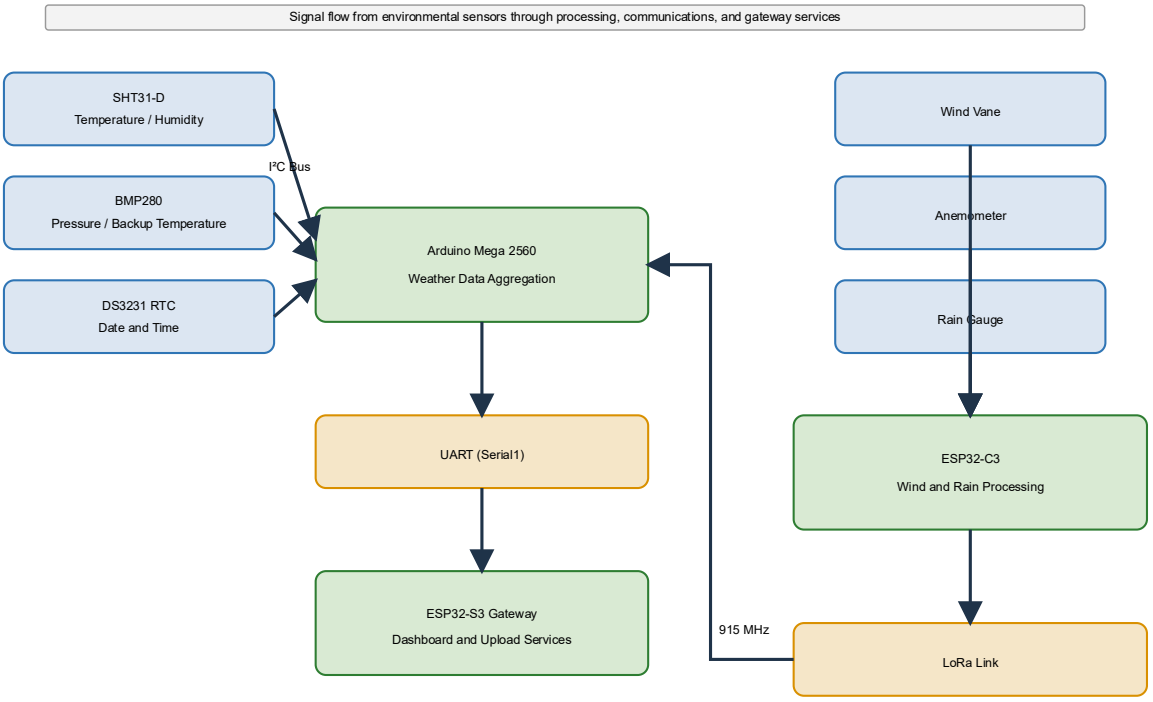
**Mast-mounted Sensors**

| Device | Interface |
|---|---|

| | |
|---|---|
| **Wind Vane** | **ADC input** |
| **Anemometer** | **Interrupt input** |
| **Rain Gauge** | **Interrupt input** |

---

## A.6 Signal Path Summary




---

**Figure A-1. Signal Path Summary**

Complete end-to-end signal flow from environmental sensors through the $I^2C$, LoRa, and UART communications layers to the Arduino Mega and ESP32-S3 controllers for processing, display, storage, and cloud distribution.

## Appendix B – EEPROM Memory Map

### B.1 Purpose

EEPROM storage preserves critical station information across power cycles.

Stored information includes:

- Environmental records
- Settings
- Reset information
- Calibration parameters

---

### B.2 EEPROM Allocation

| Address | Function |
|---|---|
| 0 | Min Temperature |
| 4 | Max Temperature |
| 8 | Min Humidity |
| 12 | Max Humidity |
| 16 | Min Pressure |
| 20 | Max Pressure |
| 24 | Altitude |
| 28–48 | Timestamp Records |
| 500 | Reset Flag |
| 520–523 | Last Reset Information |

---

### B.3 EEPROM Organization

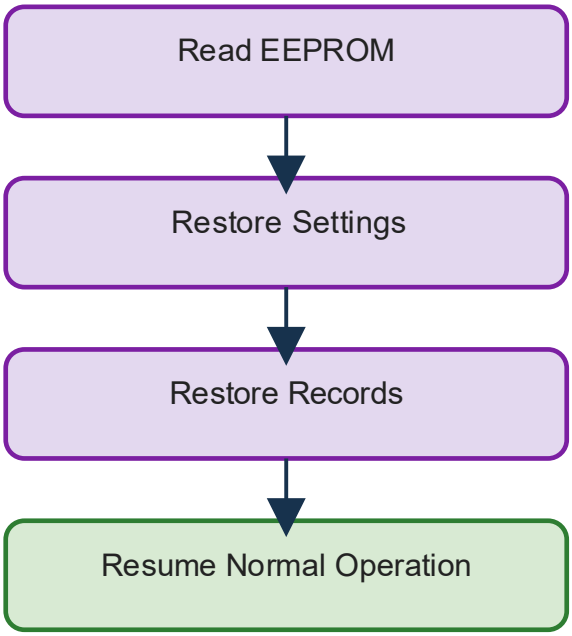
EEPROM storage is organized into four logical regions:

- **Environmental Records** store minimum and maximum weather values.
- **Timestamps** store the associated date and time information.
- **Configuration** stores persistent operating parameters such as altitude.
- **Recovery Information** stores reset and restart data used for diagnostics.

---

### B.4 Startup Recovery

Upon startup:



## Appendix C – Communications Protocol Reference

### C.1 Protocol Layers

Layer 1 LoRa

Layer 2 UART

Layer 3 HTTP

---

### C.2 OUT Packet

**Purpose**

Primary mast-mounted telemetry packet.

**Structure**

<OUT>

WindSpeed,

WindGust,

Direction,

RainRate,

DailyRain,

RainTips,

Uptime

</OUT>

---

### C.3 STATUS Protocol

**Request**

<STATUS?>

**Response**

<STATUS>

---

**C.4 REBOOT Protocol**

**Request**

<REBOOT>

**Response**

<REBOOTING>

---

**C.5 UART Weather Packet**

The Mega forwards:

Temperature

Humidity

Dew Point

Pressure

Wind Speed

Wind Direction

Rainfall

ESP32-C3 Uptime

Date

Time

to the ESP32-S3.

---

### C.6 Communications Timing

| Activity | Interval |
|---|---|
| LoRa Telemetry | ~5 sec |
| UART Update | ~5 sec |
| Status Request | ~30 sec |
| Reboot Trigger | ~60 sec |
| Retry Count | 3 |

---

## Appendix D – Database Field Dictionary

### D.1 Temperature Fields

| Field | Description |
|---|---|
| temp | Corrected temperature |
| rawtemp | Raw sensor reading |
| adjfactor | Applied adjustment |

---

### D.2 Moisture Fields

| Field | Description |
|---|---|
| humidity | Relative humidity |
| dewpt | Dew point |
| moist | Moisture classification |
| dewpot | Dew-potential classification |

---

### D.3 Pressure Fields

| Field | Description |
|---|---|
| presshpa | Pressure in hPa |
| pressinhg | Pressure in inches Hg |

---

### D.4 Wind Fields

| Field | Description |
|---|---|
| winddir | Direction |
| windspd | Wind speed |
| windgust | Peak gust |
| windlast | Previous direction value |

---

### D.5 Rain Fields

| Field | Description |
|---|---|
| rainrate | Current rain intensity |
| rainday | Daily rainfall |
| raintips | Bucket-tip count |

---

### D.6 Health Fields

| Field | Description |
|---|---|
| outuptime | ESP32-C3 uptime |
| nextupl | Upload timer |

| | |
|---|---|
| lastconn | Last connection |
| WUstat | Weather Underground status |

---

## Appendix E – System Parts List (BOM)

### E.1 Main Station

| Qty | Component |
|---|---|
| 1 | Arduino Mega 2560 |
| 1 | ESP32-S3 |
| 1 | LCD Keypad Shield |
| 1 | RYLR890 LoRa Module |

---

### E.2 Radiation Shield Sensor Package

| Qty | Component |
|---|---|
| 1 | SHT31-D |
| 1 | BMP280 |
| 1 | DS3231 RTC |

---

### E.3 Mast-mounted Sensor System

| Qty | Component |
|---|---|
| 1 | ESP32-C3 Node |
| 1 | RYLR890 LoRa Module |
| 1 | Wind Vane |

| | |
|---|---|
| 1 | Anemometer |
| 1 | Tipping-Bucket Rain Gauge |

---

**E.4 Solar Power System**

| Qty | Component |
|---|---|
| 1 | Newpowa NPA5S-12J Solar Panel |
| 1 | SOLPERK 8A Charge Controller |
| 1 | 12V LiFePO4 Battery |

---

**E.5 Spare Parts Inventory**

Recommended stock:

ESP32-C3

ESP32-S3

LoRa Module

SHT31-D

BMP280

DS3231

RJ11/RJ12 Connectors

10K and 12K Resistors (or 10K and 2K, to be combined)

Sensor Cable

---

**Appendix F – Source File Inventory**

**F.1 Firmware Files**

**Arduino Mega**

Min_Max_Temp_with_LoRa.ino

Purpose:  Main Weather Controller

---

**ESP32-S3**

ESP32_S3_Inside_Station.ino

Purpose:  Gateway / Dashboard

---

**ESP32-C3**

ESP32_C3_outside_station.ino

Purpose:  Mast-mounted Telemetry Node

---

**F.2 PHP Files**

**weatherdb.php**

Purpose:  Database Upload Handler

---

**index.php**

Purpose:  Dashboard Interface

---

**dewpoint.php**

Purpose:  Dew Point Calculations

---

**F.3 Engineering References**

New RJ11 and Wireless Connections.txt

Power Supply Info.txt

LoRa Output.txt

## Appendix G – Engineering Notes and Design Rationale

### G.1 Development Progression

The weather station evolved through several major phases:

| Phase | Major Enhancement | Result |
|---|---|---|
| 1 | Single Controller | Initial weather monitoring platform with temp and humidity sensors |
| 2 | ESP32-S3 Gateway | Network connectivity and dashboard hosting |
| 3 | Web Dashboard | Remote monitoring |
| 4 | Mast-mounted Controller | Distributed data acquisition |
| 5 | LoRa Communications | Wireless mast-mounted telemetry |
| 6 | Database Storage | Historical weather archive |
| 7 | Solar Power System | Self-powered mast-mounted operation |

---

### G.2 Major Enhancements

#### Distributed Controller Architecture

Benefits:

- Reduced software complexity
- Better fault isolation
- Improved maintainability

---

#### LoRa Communications

Benefits:

- Reduced wiring

- Improved deployment flexibility
- Better sensor placement

---

**Remote Reboot Capability**

Benefits:

- Reduced service visits
- Improved uptime
- Automated recovery

---

**Database Archiving**

Benefits:

- Historical records
- Trend analysis
- Reporting capability

---

**G.3 Why a Three-Controller Architecture Was Selected**

The weather station could have been implemented using a single microcontroller.

That approach was ultimately rejected because it would have created a controller responsible for:

- Weather measurements
- User interface
- Database communications
- Dashboard hosting
- LoRa communications
- Internet communications

By distributing responsibilities across three processors, each subsystem remains relatively independent.

**Arduino Mega**

Responsible for:

- Weather Logic
- LCD Interface
- RTC
- EEPROM
- Recovery Management

**ESP32-C3**

Responsible for:

- Mast-mounted Data Acquisition
- LoRa Communications
- Rain Processing
- Wind Processing

**ESP32-S3**

Responsible for:

- Dashboard
- Wi-Fi
- Database Uploads
- Weather Underground Uploads

This separation significantly simplifies maintenance and troubleshooting.

---

**G.4 Why LoRa Was Selected**

Several wireless technologies were considered.

**Wi-Fi**

Advantages:

- High Throughput

- Easy Network Integration

Disadvantages:

- Higher Power Consumption
- Outdoor Coverage Challenges

**Bluetooth**

Advantages:

- Low Cost
- Simple Hardware

Disadvantages:

- Limited Range

**LoRa**

Advantages:

- Excellent Range
- Low Power Consumption
- Reliable Outdoor Operation
- Simple Deployment

LoRa was selected because it provided the best combination of range and reliability for the mast-mounted sensor node.

---

**G.5 Why Solar Power Was Adopted**

The original concept assumed permanent wired power.

During development it became apparent that sensor placement and wiring flexibility would improve significantly if the mast-mounted node became self-powered.

Benefits achieved:

- No power cable
- Flexible installation
- Reduced trenching

- Improved placement options

---

**G.6 Why ESP32-C3 Uptime Was Added**

System uptime was not part of the original design.

Troubleshooting revealed that determining whether the mast-mounted node had rebooted was often difficult.

The uptime counter was introduced to provide visibility into:

- Unexpected Resets
- Power Problems
- Recovery Success
- Communications Health

It became one of the most valuable diagnostic metrics in the entire system.

---

**G.7 Why Remote Reboot Was Added**

Field failures occasionally required access to the mast-mounted node.

Because the node can be mounted outdoors and operate on solar power, physical access is less convenient than access to indoor equipment.

The remote reboot protocol was created so that:

- Communications Failures
- Software Lockups
- Temporary Faults

could often be recovered without a service visit.

---

**G.8 Why a 12K Resistor Was Selected for the Wind Vane**

Wind-Vane Signal Conditioning

The wind vane is connected to ESP32-C3 GPIO0 through a 12K pull-up resistor to the 3.3V rail.

The resistor forms a voltage divider with the vane's internal resistor network, allowing the ADC to distinguish individual wind-direction states.

A 12K pull-up resistor was selected through empirical testing. The value produced reliable separation between wind-direction ADC states while maintaining stable measurements. Testing showed that a 10K pull-up resistor did not provide sufficient separation between all vane positions.

---

**G.9 Reliability Philosophy**

The station was designed around a simple principle:  Weather acquisition should continue whenever possible.

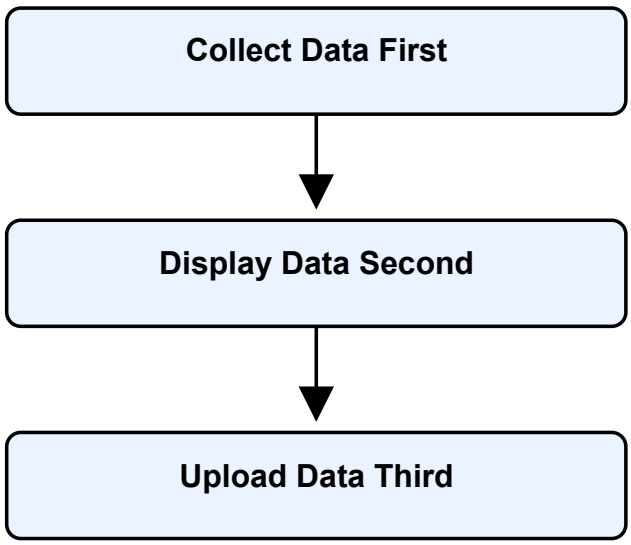



**Figure G-1. Reliability Philosophy Hierarchy**

Failures in:

- Internet Connectivity
- Database Services
- Dashboard Services

should not prevent weather collection.

This principle strongly influenced the final architecture.

---

**G.10 Design Principles**

The weather station follows the principles below.

**Simplicity**

Prefer understandable solutions over sophisticated solutions.

**Recoverability**

Subsystems should recover automatically whenever practical.

**Visibility**

Failures should be obvious.

**Documentation**

Critical behaviors should be documented.

**Expandability**

Future sensors should be easy to add.

---

**G.11 Long-Term Support Objectives**

The project was designed with an anticipated service life measured in years rather than months.

Long-term goals include:

- Minimal Maintenance
- Predictable Operation
- Simple Repair
- Incremental Enhancement

The technical manual exists partly to support these objectives.

---

**Appendix H – Future Enhancements Roadmap**

**H.1 Power Monitoring**

Future capabilities:

Battery Voltage

Charge Current

Solar Voltage

Battery Health

---

**H.2 Environmental Expansion**

**Potential additions:**

UV Index

Solar Radiation

Air Quality

---

**H.3 Dashboard Improvements**

**Potential features:**

Historical Graphs

Trend Analysis

Interactive Reporting

Mobile Optimization

---

**H.4 Communications Enhancements**

**Potential features:**

Packet Statistics

Reliability Metrics

Signal-Quality Tracking

Advanced Diagnostics

---

**H.5 Remote Administration**

**Future possibilities:**

Remote Configuration

Remote Calibration

Remote Firmware Updates

---

**H.6 Version 2.0 Vision**

Potential Version 2.0 capabilities:

Solar Telemetry

Battery Telemetry

Historical Charting

Advanced Forecasting

Remote Updates

Additional Sensors

Enhanced Analytics

while maintaining the proven architecture:

Arduino Mega

ESP32-S3 Gateway

ESP32-C3 Mast-mounted Node

LoRa Communications

Dashboard

Database

---